

TRADE AND THE END OF ANTIQUITY

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How did trade and economic activity evolve at the End of Antiquity, when political power shifts away from the Mediterranean towards the Middle East and northern Europe? No records of ancient production, consumption, and trade exist. Yet the composition of coin finds reveals information about monetized transactions, as coins gradually diffused along trade routes. To recover trade and consumption time series, we assemble a database of half a million ancient coins from the fourth to the tenth century and estimate a dynamic model of trade and money. Our estimates suggest that: Mediterranean trade was disrupted by the newly formed border between Islam and Christianity; economic activity shifts away from the Mediterranean starting in the sixth century; real consumption peaks in the Middle East in the eighth century; and by the end of the ninth century, Islamic Spain and Frankish northwestern Europe are the wealthiest regions of the ancient western world.

KEYWORDS: Gravity Equation, International Trade, Ancient Coins, Diffusion.

1. INTRODUCTION

The transition from classical antiquity to the medieval period marks one of humanity's most profound transformations, altering the intellectual, economic, and political fabric of Europe, the Mediterranean, and the Middle East. This sweeping metamorphosis reshaped entire economic systems and political structures, transforming a world organized around Mediterranean commerce and urban networks into one dominated by new centers of power. We present new data and empirically document this changing economic geography during Late Antiquity. This quantitative analysis sheds new light on the nature and extent of economic change as the Roman civilization in the Mediterranean declines and economic power shifts to the Middle East and to western and northwestern Europe.

What caused the end of antiquity has been a central question for centuries (see for instance [Montesquieu, 1734](#), [Voltaire, 1756](#), [Gibbon, 1789](#)). Most contemporary historians believe that the conquest of Rome by Germanic invaders in the fifth century did not lead to an immediate end of Roman institutions which remained largely in place ([Pirenne, 1927, 1939](#), [Findlay and O'Rourke, 2009](#), [McCormick, 2001](#)).¹ The archaeological evidence points to a shift in the economic activity to the north-west of Europe and away from the Mediterranean between the fifth and the eighth century. The timing, extent, and reasons remain debated, but when Charlemagne was crowned as Emperor at the end of the eighth century, the political and economic power in Europe has moved from the Mediterranean to the Frankish lands of northwestern Europe. Famously, historian Henri Pirenne proposes that the expansion of the Arab Caliphate along the

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¹ *Contra*: [Ward-Perkins \(2006\)](#).

southern Mediterranean coast and into the Iberian peninsula disrupted commerce and political ties in the Mediterranean, and turned the emerging Carolingian Empire into a northern European power (“without Mohammed Charlemagne would have been inconceivable,” [Pirenne, 1927](#), p.234). The evidence for these disruptions brought forward by Pirenne is mainly related to the disappearance of certain luxury goods north of the Mediterranean.²

In this paper, we study the changing economic geography during Late Antiquity using the tools of modern quantitative trade models and novel data on the circulation of ancient coins. Our aim is not to *explain* the end of antiquity, but to present new quantitative evidence to *describe* it. Our evidence suggests that the emergence of the Arab Caliphate reshaped Mediterranean trade, with north-south trade collapsing while south-south trade flourished. But we also show that the relative economic decline of the Mediterranean world started more than a century earlier; and that the Atlantic regions from the Islamic Iberian peninsula to the Frankish lands of northwest Europe became the wealthiest regions of the ancient western world more than seven centuries before the European colonial expansion into the Americas, Africa, and Asia.

We make four main contributions. Our first contribution is to assemble a large database of coin finds from hoards deposited between AD 325 and AD 950, with observations from hundreds of thousands of coins found across Europe, North Africa, and the Middle East. Coins offer rich quantitative information in a data-scarce setting,³ as numismatists and archaeologists have deciphered, cataloged, and classified ancient coinage for over 200 years. We collect information about where and when coins were minted and buried, and present three stylized facts: (i) bilateral coin flows are disrupted by distance and political borders just like trade flows are, (ii) unlike traded goods, coins are in use over many years and, as a result, coins that are older at the time of deposit tend to have traveled farther, and (iii) the geography of coin flows across the Mediterranean changes abruptly around the time of the Arab conquests of the 7th century.

Our second contribution is to build and estimate a dynamic model of trade where agents use coins for transactions. Within each period, trade is governed by comparative advantages as in [Eaton and Kortum \(2002\)](#), and coins flow in opposite direction to trade. After being minted, the same coin can then be saved as a store of value, and re-used for subsequent transactions. We first show that with saving, coin flows within a period inherit the same gravity structure as trade flows, up to a single multiplicative constant. We then characterize the full dynamics of coin flows, as coins are minted, saved, used for multiple transactions, and gradually percolate through the trade network. Even though individual coins only contain information about their initial place and time of “birth” (minting) and ultimate “death” (deposit) but leave no trace of any of the intermediate transactions they were used for, we show that data on overlapping generations of coins is sufficient to fully recover the parameters governing coin creation (minting) and trade in goods within *each period* (trade costs and production capacity).⁴ Our estimates for minting output are in line with known historical evidence, and our estimate for the travel time elasticity of trade is similar to that for Roman trade in ceramics ([FHLL+, 2022](#)), and

²Pirenne’s argument is the near *absence* of mentions of silk and spices in historical texts written north of the Mediterranean, and the disuse of gold for coinage and papyrus for writings. These fragments of evidence, along with new archaeological findings, have been extensively studied and discussed by historians since Pirenne. See [Lopez \(1943\)](#), [Ashtor \(1970\)](#), [Hodges and Whitehouse \(1983\)](#), and, in particular, [McCormick \(2001\)](#)’s monumental work that synthesizes the existing literary and archaeological evidence on changes around the Mediterranean, including patterns of change in the flows of communications, objects, and travelers. The synthesis of [Wickham \(2006\)](#) interprets the evidence through the lens of social structures.

³Due to the fact that no comprehensive data on production, consumption, trade, or demographics exists for the first millennium AD, historians of this period rely to a large extent on qualitative literary sources.

⁴As recognized by numismatists, whether a coin hoard is created (deliberately or accidentally buried), found by archaeologists, and documented by numismatists, depends on a series of endogenous events. To purge our estimation from those endogenous events we use only information on the *shares* of different coins within a hoard.

Bronze Age Assyrian trade in textiles and precious metals (Barjamovic, Chaney, Coşar, and Hortaçsu, 2019). Our estimates reveal a large cost associated with crossing the newly erected border between Islamic and non-Islamic regions, especially over the Mediterranean.

Our third contribution is to reconstruct granular time series for bilateral trade flows and for real consumption for each 20-year period from the 4th to the 10th century for each region. We are able to recover relative real consumption using solely data on coins because coin flows contain information on (nominal) trade flows, which contain information on (relative) prices. Our estimates confirm the thesis proposed by Pirenne (1939) that north-south Mediterranean trade collapses after the Arab conquests in the 7th century, although we also show that south-south trade remains vibrant. Our estimates further reveal that Mediterranean income falls relative to the rest of the ancient western world starting in the 6th century, and that the concentration of income falls sharply in the 5th century and remains stable thereafter. Our estimates also suggest that real consumption in the heartlands of the Byzantine empire, initially among the highest, gradually collapses starting in the 7th century, in part due to the fall of trade in and out of regions newly conquered by the Arabs, while consumption experiences a spectacular rise in northwest Europe in the 7th and 8th centuries; consumption in the heart of the western Roman empire regions of Italy, southern France, and Egypt starts falling in the 4th century, around the time when the institutions of the western Roman empire collapse; and consumption peaks in Syria under the Umayyad Caliphate of Damascus in the early 8th century, in al-Andalus under the Umayyad Caliphate of Cordoba in the late 8th century, in al-Iraq under the Abbasid Caliphate of Baghdad, and in the Arabian peninsula in the 8th century after the birth of Islam. Aggregate consumption rises in all Atlantic regions from North Africa to northwest Europe starting in the 8th century, more than seven centuries before the European colonial expansion.

Our fourth contribution is to confront our estimates to a series of external data. The known exploitation of ancient mines is positively associated with our estimates of coin minting. Adverse paleo-climatic shocks — severe droughts — have a negative impact on real consumption, but military conflicts and the local incidence of the Justinianic plague do not. Finally, in the absence of virtually any systematic evidence on ancient production, consumption, or trade, we show that our estimates on real consumption changes from pre- to post-AD 700 are remarkably consistent with measures of European urbanization post-AD 700.

Related literature. Our paper relates to the literature on the role of market access in shaping economic outcomes across space, specifically in historical settings. Fogel (1964), Donaldson and Hornbeck (2016) and Hornbeck and Rotemberg (2024) evaluate the impact of the US railroad in the 19th century on economic growth, Donaldson (2018) the impact of railroads in colonial India on relative welfare, and Nagy (2023) the impact of the westward expansion of the US on growth; Pascali (2017) evaluates the impact of steamships on maritime trade and relative development; Redding and Sturm (2008) study the impact of the iron curtain on comparative development in Germany, and Ahlfeldt, Redding, Sturm, and Wolf (2015) the impact of the Berlin wall on the urban structure of Berlin; Juhász, Sakabe, and Weinstein (2026) study the impact of knowledge codification on industrial development in Japan, using trade data to infer aggregate productivity; FHLL+ (2022) study the impact of the Roman transportation network on trade in ceramics (terra sigillata) from the 1st century BC to the 3rd century AD, while Michaels and Rauch (2018) study changes in the transportation and urban networks after the fall of the Western Roman Empire; Barjamovic, Chaney, Coşar, and Hortaçsu (2019) use shipment records from Assyrian merchant archives in Bronze Age Anatolia to estimate the location of ancient lost cities and their size. In contrast to this literature, we do not observe prices, trade costs, or even trade flows. Instead, we use a dynamic model of trade and money to recover trade flows from panel data on the movement of coins. Liu and Tsyvinski (2024) feature a related

dynamic model where shocks gradually percolate through an input-output network. We follow an approach similar to [Donaldson \(2018\)](#) and [Donaldson and Hornbeck \(2016\)](#) to measure travel time — optimal routing algorithm on a transportation network, but we add substantial historical evidence ([Scheidel, 2015](#), [Romanov and Seydi, 2022](#), [Cornu, 1983](#)) including direct evidence from 10th-century Arab geographer [Al-Muqaddasī \(985\)](#). Finally our paper speaks to a literature in economic history on the changes in Late Antiquity and early medieval times. This literature frequently uses numismatic evidence, although mostly in a descriptive manner. Two notable exceptions are [Noonan \(1980\)](#)’s study of Islamic coin finds of different vintages and origins in Eastern Europe and Scandinavia, which stops short of using a formal econometric model; and [Persson and Sharp \(2015\)](#) who discuss Pirenne’s thesis and economic integration in Europe through the lens of a gravity model.⁵

The remainder of the paper is structured as follows. Section 2 lays out the historical context and describes our data and three stylized facts on ancient coin flows. Section 3 presents our dynamic model of trade and money. Section 4 describes our estimation method and presents our estimates. Section 5 uncovers novel facts on trade, consumption, and the end of antiquity. Section 6 confronts our estimates to external data on ancient mines, climate, urbanization, battles, and plagues. Section 7 concludes. The Supplemental Appendix contains detailed information on data provenance and assembly, and additional empirical and theoretical results.

2. HISTORICAL CONTEXT, DATA, AND STYLIZED FACTS

2.1. *Historical context*

We study the economic and political developments in the Mediterranean between the 4th and the 10th century AD. At the start of this period the Mediterranean was still entirely under the control of the Roman Empire, albeit at times with multiple emperors and conflict between them, and under mounting pressure from Germanic invasions. The death of the eastern emperor Theodosius I in 395 divided the Roman Empire into a western and an eastern half. The fifth century saw increased Germanic incursions in the east and west, culminating with the Ostrogothic king Odoacer deposing the last West Roman emperor Romulus Augustulus in 476 and ending the Western Roman Empire. Italy was ruled by the Ostrogoths until the 550s, then by the Lombards; Spain was taken over by the Visigoths, France by the Merovingian dynasty of the Franks, and North Africa, Sicily, and Sardinia by the Vandals. The Eastern Roman (Byzantine) Empire at times reconquered parts of the former Western Roman territory, but in the sixth century became increasingly under pressure from the Sasanian (Persian) Empire in the east. The Byzantine-Sasanian wars of 602–628 depleted the resources of both empires, leaving a vacuum that was filled by the emerging Arab caliphate following the birth of Islam.

Figure 1 shows the rapid Arab expansion, starting in 632. By 634 the Arabs controlled the entire Arabian Peninsula. The Levant followed in the late 630’s and Egypt in the 640’s. By the end of the Rashidun Caliphate in 661, the Arabs controlled land from Tripoli in the west to Balkh in the east. The expansion continued under the Umayyad dynasty. In 698 the Arab army razed Carthage and by 709 they had fully conquered the Maghreb. In 711 they crossed the Strait of Gibraltar and defeated the Visigoths at the Battle of Guadalete. In 732 they were stopped by Charles Martel at Tours, and driven back across the Pyrenees. When the Abbasid family overthrew the ruling Umayyads in 750, the Umayyads retained control of most of Iberia (al-Andalus). While the Arab conquest ended Sasanian rule in the east, advances into Byzantine territory in Anatolia did not lead to sustained land border shifts. But the Arabs strengthened their naval capabilities, ended the Byzantine control of the western Mediterranean and contested its control of the east. Arab raids on Mediterranean cities became frequent.

⁵[Shatzmiller \(2018\)](#) qualitatively applies the gravity model to medieval trade in the Middle East.

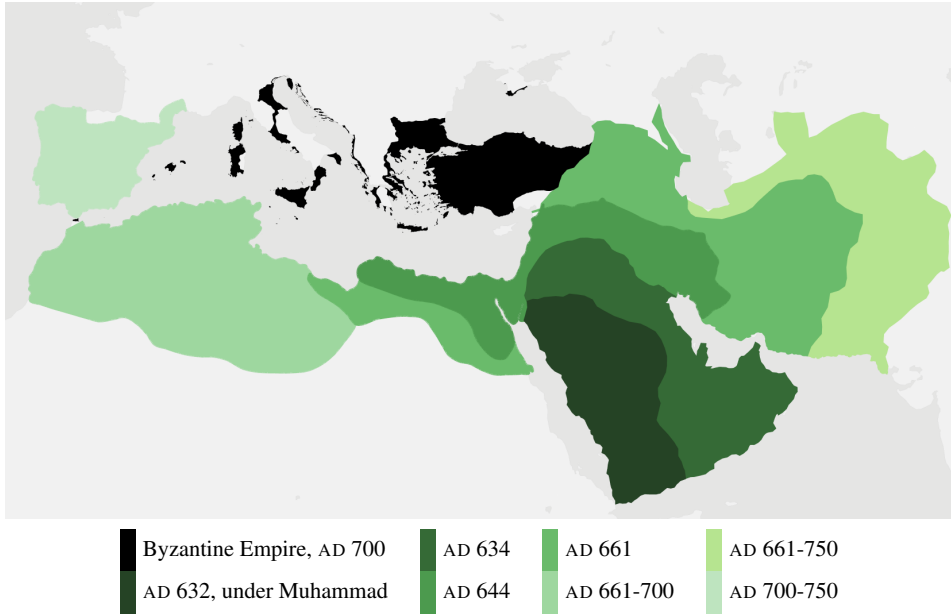


FIGURE 1.—The Arab Conquests, AD 632-750

Two other notable and overlapping events have been linked to economic and political changes: the Plague of Justinian (541-549), which, according to contemporary literary sources led to large declines in population, and the “Late Antique Little Ice Age” temperature anomaly (536-560), likely due to volcanic eruptions, which caused temperatures in the northern hemisphere to drop by about one degree Celsius (Peregrine, 2020). The size and quantitative relevance of these two events is heavily debated among historians. We will think of these events as potentially affecting population and productivity, which our estimation strategy accounts for.

2.2. Data

We construct a large dataset on the flows of coins around the Mediterranean between AD 325 and AD 950.⁶ For the period from AD 325 to AD 725 we mostly rely on data from the *Framing the Late Antique and Early Medieval Economy* project (FLAME, 2023b),⁷ a large-scale effort by historians and numismatists to record harmonized information on the location, dating, and composition of coin finds up to the year 725. FLAME covers hoards⁸ from the Mediterranean and beyond, contributed by specialists working on the coinage of their geographical and temporal expertise. We use the most recent release of FLAME (January 2025) which covers 9,831 coin hoards. We remove hoards that fall outside our area of interest, continental western Europe up to the modern-day German-Polish border and including Bohemia, southern Europe up to the line between Vienna and Odessa, and North Africa and the Middle East up to the maximum extent and area of influence of the Umayyad Caliphate (stretching from the Maghreb in the

⁶See Supplemental Appendix D for extensive information on the coin data assembly, harmonization and cleaning.

⁷<https://coinage.princeton.edu/>

⁸FLAME also includes finds from excavations and single finds. Unless explicitly mentioned, we will treat all records in the same way and just use the word “hoard” to describe deposits of any size.



FIGURE 2.—Ancient coin, example. *Notes:* Both sides of an Islamic gold dinar, diam. 1.9cm, wt. 2.8g, found in Nishapur, Iran (findspot). The central inscription on the left panel mentions the reigning Samanid ruler, Nuh bin Nasr, and the inner margin states “In the name of God, this dinar was struck at al-Muhammadiyya in the year three hundred and thirty-three.” Those marks indicate the mint location (al-Muhammadiyya, near the modern city of Rayy, Iran) and the mint date (333 of the Hijri calendar, AD 944–45). *Source:* Metropolitan Museum of Art, Obj. Nr. 39.40.127.557.

West to the Indus in the east, and up to Bulgar in the north).⁹ We also remove all hoards that only consist of incompletely described coins (no mint location or mint date information).

We supplement FLAME’s data, in particular for the period after AD 725, with hand-coded records of 106,356 coins from 829 finds, which we assemble using hoard catalogues from the numismatic literature, similarly to the source documents that underlie FLAME. These additional records include the time period of the Caliphate, so that we can assess the impact of the birth of Islam on the patterns of exchange in the Mediterranean. Together these data cover the vast majority of published information on coin finds in our geographic and temporal scope.

The structure of the coin hoard data is ideally suited for an analysis of dynamic bilateral spatial flows. Each unit of observation—a coin within a hoard—contains the following attributes: (i) the location where the coin has been minted, (ii) the ruler and/or dynasty under whose authority the coin has been issued, (iii) a year interval when the coin was minted, (iv) the identifier and the location of the hoard that the coin is part of, (v) a year interval when the hoard—and therefore the coin—was deposited. These pieces of information are typically recorded by the author of the original numismatic or archaeological publication. Mints are usually inferred from mint marks on the coins.¹⁰ The mint date is often indicated on the coin. When the date is not indicated, it can be approximated from the ruler (or dynasty or empire) under whose authority the coin was issued, as well as other information, including the mint mark. Figure 2 displays a well preserved example of an Islamic coin. Finally, we follow the common approach of historians to estimate the date of deposit of each hoard using the *terminus post quem*, or *tpq* for short, the date of the youngest object in the hoard that can be dated. In our case that is typically the most recent end year of the time intervals of the coins in the hoard.

⁹See figure 6 below for a map of our area of study. We exclude the Viking lands, and do not speak to the discussion on the potential role of the Vikings (and the inflow of Islamic silver through trade via eastern and northern Europe) in the changing economic geography during Late Antiquity (Bolin, 1953). Recent archaeometric studies indicate that Carolingian silver is largely not of Arab origin (SBMR+, 2008, Kershaw, Merkel, D’Imporzano, and Naismith, 2024), suggesting a limited role for silver inflows via the Viking route in affecting Carolingian mint output.

¹⁰Mint marks have been in use since ancient Greek times to monitor coin weight and metal content.

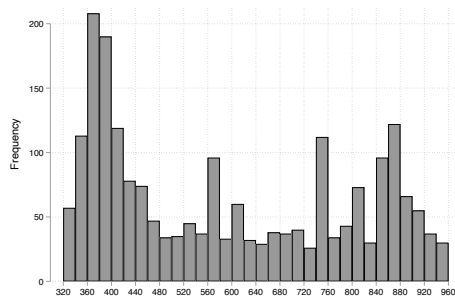
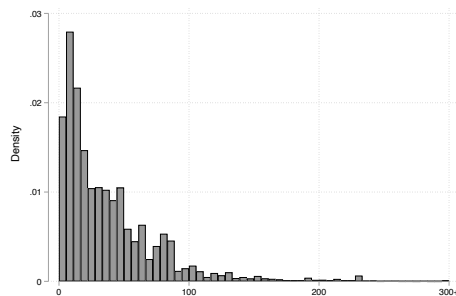
(a) Distribution of hoards by *tpq*, AD 325-950(b) Coin age at time of deposit (*tpq*), in years

FIGURE 3.—Number of hoards over time and ages of coins within hoards. *Notes:* Panel (a) shows the number of hoards per 20-year period. Panel (b) displays the (annual) density of coins of different ages. Coin age is defined as the difference between the midpoint of the coin's minting interval and the *tpq* of the hoard it is found in.

In coding the mint location and date we typically follow the coding of the author of the original publication which catalogues the hoard.¹¹ In some cases this information is imprecise: the author may not have been able to inspect the coin or inspected only a fragment. We conduct robustness checks to investigate whether our findings are driven by endogenous selection.

We have data on 5,501 hoards and 491,432 coins that fall into our geographic boundary and have a *tpq* between 325 and 950. After removing from FLAME large hoards found in the 19th century or earlier for which not much besides rough coin counts is known, 268,209 coins are complete with a mint and minting year interval; on average 87% of coins in a hoard are complete. We define the age of coins at time of deposit as the difference between the midpoint of the coin's minting interval and the *tpq* of the hoard. Figure 3a shows the temporal distribution of the number of hoards by *tpq*, and figure 3b the distribution of coin ages within hoards.¹²

Coins are deposited on average 59 years after they are struck. Supplemental Appendix tables B.I and B.II contain additional summary statistics on coins and hoards respectively.

Discussion. The interpretation of coin flows as relating to trade, despite having a long tradition among numismatists and historians,¹³ deserves some discussion. The Roman and subsequently Byzantine empires were generally fairly monetized economies,¹⁴ with coinage taking a pre-eminent role and credit being very limited (Morrisson, 2002). The situation was similar in the caliphate (Bessard, 2020) and in the Carolingian empire (Coupland, 2014). The fact that coins were light, durable, and—because they were made from precious metal, which could be easily reminted—accepted within and across borders made them particularly suitable for long-distance trade.¹⁵ This is particularly the case for gold coins, traded throughout the Mediter-

¹¹Sometimes these interpretations are critically evaluated and corrected by subsequent scholars. Supplemental Appendix D lists the extensive sources we use for each of the hand-coded hoards.

¹²To further validate that our data reflect coin circulation during Late Antiquity, Supplemental Appendix B.I compares the coin age distribution with the one in twelve Byzantine hoards that Banaji (2016) labels as circulation hoards, i.e. that are known to have originated from coins used for trade transactions. Coin age distributions are similar.

¹³See, in particular, the discussions by Grierson (1959) and, more recently, Naismith (2014).

¹⁴A possible exception was the eighth century, where Byzantine mint output collapsed.

¹⁵Examples of coins used in foreign empires abound. Bates (1991) discusses how Byzantine coins kept circulating (and being minted) in Egypt after the Arab takeover in 641. Tribute and ransom payments between the Arabs and Byzantines often included domestic currency. McCormick (2001), chapter 12, discusses the circulation of Arab and Byzantine coins in the west.

anean and valued for their weight in gold (Banaji, 2016).¹⁶ Of course coins did not travel solely because of commerce; theft, gift-exchange, dowry, tribute, ransom and plunder also contribute to coin flows. We acknowledge that while tribute and ransom can be thought of as the price paid for life, theft and plunder may not easily fit into our trade model.¹⁷

A potential source of bias comes from the fact that our data do not cover the universe of coin flows, but only hoards that have been created (coins deliberately or accidentally deposited, which may depend on warfare, natural disasters, and property rights protection), found (which may depend on modern-day institutions, such as whether metal detecting is allowed, and on modern-day market prices for historical coins), and documented by experts (which may depend on the local presence of experts, the ‘novelty’ of the hoards’ contents, and the demand for research on these topics).¹⁸ Our estimation in section 4 corrects those biases by using shares of coins *within* hoards, and differs from the more descriptive methods employed by historians.

Finally, in contrast to standard trade data, we do not observe flows at each point in time, but only when and where a coin was minted and eventually deposited into a hoard. Our structural model in section 3 disentangles the possibly numerous successive trips a coin took throughout its life, and identifies from coin data the parameters governing trade flows each period.

2.3. Three stylized facts on ancient coins

Distance and political borders disrupt coin flows. The bilateral structure of our dataset allows us to explore the geography of coin flows in reduced form. We aggregate mint-origin (m) and hoard-destination (h) coordinates to $1^\circ \times 1^\circ$ cells across all periods, and restrict the sample to coins that were unambiguously issued by the fourteen most important political blocks p .¹⁹ We then model the number of coins issued by p in m and found in h as a function of distance and a political border dummy,

$$\text{count}_{mhp} = \exp(a_{mp} + a_h + b_1 \ln(\text{distance}_{mh}) + b_2 \text{PoliticalBorder}_{hp} + u_{mhp}). \quad (1)$$

We estimate this model by PPML using data on all triplets (m, h, p) for mint cell m in political block p and hoard cell h , excluding combinations of mint cells and political blocks that do not produce a single coin. The political border dummy is one if the region where the center of the hoard cell h is located in has never and to no extent been under the political control of p .²⁰

Table I shows that distance and crossing a political border are both negatively associated with coin flows.²¹ Columns 3 and 4 show almost identical results when using only gold coins, only 6% of the coins in our sample, which were universally valued for their metal content and

¹⁶We conduct robustness checks of our main results restricting the sample to gold coins.

¹⁷Other data-driven approaches used by economic historians to measure economic activity include urbanization rates, flows of consumption goods, notably ceramics (Wickham, 2006, FHLL+, 2022), communication flows and movements of people (McCormick, 2001), pollen grain measurements (Izdebski, Koloch, Słoczyński, and Tycner, 2016), and ice core readings (MWSA+, 2018). Coin flows bring several econometric advantages and are more directly related to comprehensive patterns of exchange than communications or ceramics. We nonetheless estimate similar distance elasticities from gravity regressions on coins and ceramics (see Supplemental Appendix B.2 and figure B.2).

¹⁸FLAME (2023a) discusses potential sources of biases, which also apply to our combined data.

¹⁹These are: Eastern Roman Empire, Roman Empire, Umayyads of Spain, Abbasids, Umayyads, Sasanians, Fatimids, Western Roman Empire, Carolingians, Samanids, Visigoths, Ostrogoths, Merovingians, Vandals.

²⁰See Supplemental Appendix A.3 for our division of the Arab and Mediterranean world into 21 regions. In our structural analysis, section 4, we use 13 coarser regions. See figure 6 for a map.

²¹An alternative explanation for the border effect could be that coins are redistributed within a political entity before entering circulation. Supplemental Appendix table B.III shows similar results for specifications with hoard $\times$ political block fixed effects, suggesting that this is unlikely.

TABLE I
DISTANCE AND BORDER EFFECTS IN COIN FLOWS

	Dependent variable: # Coins _{mdh}				Dep. var.: Value _{mdh}	
	(1)	(2)	(3)	(4)	(5)	(6)
Log Distance	-1.140 (0.13)	-0.958 (0.14)	-1.116 (0.097)	-0.973 (0.084)	-1.138 (0.082)	-0.983 (0.070)
Political border		-2.598 (0.38)		-1.664 (0.31)		-1.462 (0.25)
Hoard Cell FE	Yes	Yes	Yes	Yes	Yes	Yes
Mint $\times$ Empire Cell FE	Yes	Yes	Yes	Yes	Yes	Yes
Sample			Gold only	Gold only	Gold and Silver	Gold and Silver
Pseudo- R^2	0.764	0.782	0.806	0.815	0.806	0.815
Observations	202132	202132	43821	43821	133451	133451

Note: This table presents estimates of various specifications of equation (1). The dependent variable is the number of coins in a hoard cell h from a mint cell m issued by a political entity p in columns 1-4, and the value of those coins in columns 5-6. The regression drops all (m, p) combinations that have no emitted coins, as well as singletons and separated observations. Hoard and mint cells are $1^\circ \times 1^\circ$. Political entities are categorized into fourteen divisions. Columns 1-2 use data on all coins, columns 3-4 gold only, and columns 5-6 value-weighted gold and silver. Standard errors in parentheses, clustered at mint cell $\times$ empire and hoard cell level.

were therefore particularly favored for long-distance trade. Column 5 and 6 also show similar results when using silver and gold coins and weighing them by their relative value.²²

Those results suggest that coin flows contain information related to trade costs (e.g. distance and border effects).²³ Our model in section 3 isolates features of the geography of coin flows that are driven by trade.

Older coins travel further. Coins are found, on average, in hoards 800 kilometers from their mint. But within hoards, older coins are also coins that have on average travelled farther. Table II shows results from a regression of log distance between a coin's mint and hoard place on log coin age — the difference between minting and the hoard's tpq — with hoard fixed effects to isolate within-hoard variation. The coefficient on coin age is positive and significant, even when including mint $\times$ mint time fixed effects to control for the average distance traveled and age of coins of a particular mint and issue. This suggests that older coins have been used on average for more transactions, each taking them further away from their mint origin. Our model in section 3 disentangles the many transactions of older coins from the few of younger coins.

The geography of coin flows changes sharply around the Arab conquests. Finally we show descriptives of the movements of coins during Late Antiquity. Figure 4 illustrates the changes in the flow of coins across the Mediterranean before and after the Arab conquests. Panel 4a shows

²²We calculate the equivalent gold weight of silver coins in two steps. First, we code the reference weights of coins of different denominations in our data, noting that coins are often clipped, broken, debased, or abraded (Manas and Velde, 2021). Second, we convert this reference weight into a gold-equivalent weight assuming a constant conversion ratio of 12g of silver for 1g of gold, a rough approximation given the fluctuations of the gold-silver ratio between 1:10 and 1:16 (Bolin, 1953). Gold represents 80% of the resulting value in our data. Since the price of copper/bronze fluctuates heavily during Late Antiquity (see Banaji, 2016, Ch. 5) and copper denominations frequently traded at values different from their intrinsic metal content, we only use silver and gold coins. We also note that FLAME does not record coin weights, resulting in approximate calculations based on official denominations.

²³To investigate the informativeness of coins for trade flows, Supplemental Appendix figure B.2 compares the distance elasticities for coins and for Roman Terra Sigillata ceramics, the only tradable good from Antiquity on which substantial amounts of data are available (FHLL+, 2021). We find similar but slightly higher distance elasticities for Terra Sigillata, possibly due to coins being more frequently re-used (see section 3.2).

TABLE II
COIN AGE AND DISTANCE TRAVELED

	Dependent variable: Log Distance between Mint and Hoard				
	(1)	(2)	(3)	(4)	(5)
Log Age of Coin	0.110 (0.039)	0.0882 (0.027)	0.0809 (0.032)	0.120 (0.035)	0.0544 (0.021)
Sample	No non-hoards No non-hoards				
Hoard FE	Yes	Yes	Yes	Yes	Yes
Mint × 50-year-interval FE		Yes			
Mint × 25-year-interval FE			Yes		Yes
R^2	0.743	0.843	0.850	0.744	0.873
Observations	263606	263411	263264	226556	226261

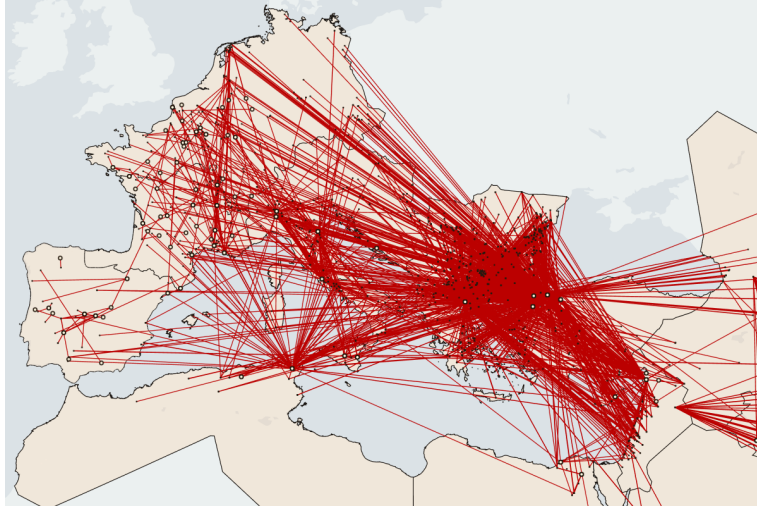
Note: The dependent variable is the log distance between the mint and hoard locations. The independent variable is the log age of the coin at the date of the *tpq* of the hoard (where the age is defined as the difference between the midpoint of the minting interval and the maximum of the endpoints of the minting intervals). In the rare cases where this age is zero (the youngest coin in the hoard is dated to a precise year) we set the log age to zero. Mints are identified as all Nomisma or FLAME-recorded entities that have been geocoded to the same $0.1^\circ \times 0.1^\circ$ cell. Columns 4 and 5 exclude FLAME finds that are tagged as not being hoards. Standard errors in parentheses, clustered at the hoard level.

flows from AD 450 to AD 630. Constantinople, Thessalonica, Rome, Ravenna, and Carthage are important mints whose coins flow across the entire Mediterranean. Coins from Carthage cross the sea into Europe, and coins from Rome and Constantinople cross into Africa and the Middle East. Panel 4b shows that the patterns of coin flows are starkly different after AD 713, by which time the Arabs have conquered the eastern Mediterranean coast (up to and including Antioch), the southern Mediterranean coast, and most of the Iberian peninsula. Most coins now flow east to west within the Islamic Caliphate, within the Arab heartlands of Syria, Mesopotamia, and Egypt, and within the Frankish lands of northern Europe. The coin flows emanating from the remaining Byzantine mints in Constantinople, Syracuse, and Italy are much smaller than in the earlier period.²⁴ Besides a few coins from the mints of Ifriqiya and al-Abbasiyya that end up in the hoards of Ilanz and Steckborn (McCormick, 2001), there are almost no north-south flows across the Mediterranean. Flows that cross the border between Christianity and Islam primarily do so in the West across the Pyrenees (Parv  rie, 2014). Supplemental Appendix B.4, figure B.3, and table B.IV present additional evidence that coin flows across the Mediterranean dropped after the Arab conquests, and that Islamic coins crossing the sea replace Roman coins. Despite the fact that in some historical sources the Arabs called the Mediterranean the “Sea of the Romans” (*Ba  r al-R  m  i*), after the Arab conquests the Mediterranean became, at least when it comes to coin flows, an Arab-dominated sea.²⁵

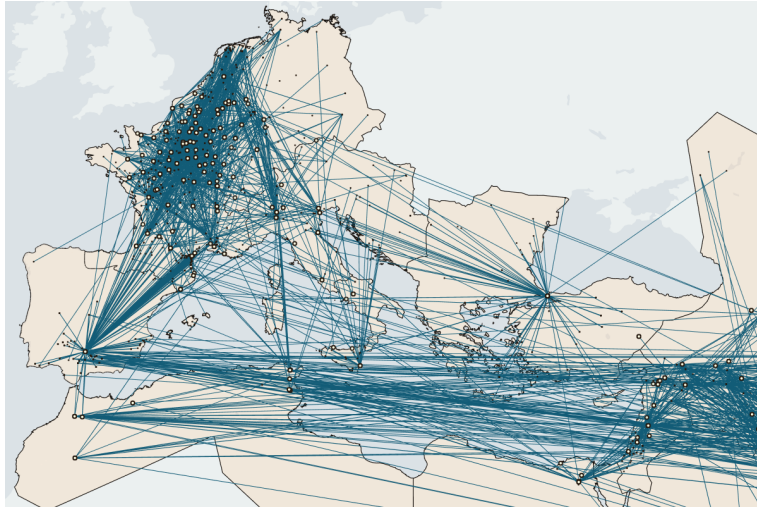
While only descriptive, figure 4 suggests that the Arab conquest coincided with a substantial change in the economic geography of the ancient world. The structural model in the next section is designed to quantify those changes.

²⁴The changes in the magnitude and location of Byzantine coin production has been the topic of a large literature. Kazhdan (1954) was the first to argue for a decline of Byzantine cities in the 8th and 9th century based on archaeological evidence. Several authors (including Kazhdan, Zavagno, 2022, and Pennas, 1996) relate these changes to Arab military pressure. Grierson (1973) notes that the eastern mints of Nicomedia, Cyzicus, Thessalonica, Cyprus, as well as Catania, were closed in 629-630, before the Arab conquests, and production was relocated to Constantinople. Nevertheless, a number of provincial mints, including Syracuse, Ravenna, and Rome, remained active until at least the mid-8th century (in the case of Syracuse until 878 when it fell to the Aghlabids).

²⁵Paraphrasing Pirenne (1939). We take the Arab conquests as a proximate cause for these changes, and do not attempt to explain why the Arabs were successful. The commonly held view is that the Byzantine-Sasanian war of 602-628 exhausted the forces of both empires and paved the way for Arab military success (Foss, 1975).



(a) Before the Arab conquests: AD 450-630



(b) After the Arab conquests: AD 713-900

FIGURE 4.—Changes in coin flows in the ancient world. *Notes:* The figure shows coin flows, indicated by a straight line, between mints (hollow circles) and find spots (black dots). The sample consists of all coin groups where both the lower end of the mint interval and the *tpq* of the hoard lie between AD 450 and AD 630 (panel a) and AD 713 and AD 900 (panel b). Hoards from outside the shaded area are excluded.

3. MODEL

We now introduce a quantitative model of trade, money, and the diffusion of coins. This model forms the basis for the estimation of trade costs, mint outputs, and technology and labor, from which we quantify changes in the economic geography of the ancient world.

We make two key assumptions. The first is that coins are used for transactions. This assumption is necessary for identification: if coins were not used to clear gross bilateral trades, then data on coins would not contain information on bilateral trade flows. The use of coinage was widespread: as historians have noted (e.g. [Mayhew, 2019](#)), the presence of millions of coins

in present-day collections (only a small fraction of which can be associated to a hoard and included in our sample) indicates that great quantities of coinage must have circulated at the time.²⁶ The period we study does not yet feature the widespread use of financial contracts for trade so that coins are the primary medium of exchange. Finally, our finding that coin movements exhibit similar empirical patterns as trade flows also supports this assumption.²⁷

Our second key assumption is that coins are fungible. This assumption is natural with coins being valued for their precious metal content, and is supported by historical evidence on the wide circulation of foreign coins and the discovery of precise tools for measuring the weight and fineness of foreign coins (Bates, 1991, McCormick, 2001).²⁸

3.1. Economic model

Set up. There are N locations, indexed by a subscript. Time is discrete, indexed by t in square brackets. Each time period is decomposed into three sub-periods: beginning, middle, and end.

At the end of period $t - 1$, location n sets aside $S_n[t]$ coins for consumption and saving in period t . At the beginning of period t , an exogenous fraction $\lambda_n[t]$ of the coins in this stock $S_n[t]$ ceases to circulate, either lost or melted into fresh new coins.²⁹ In addition, some locations own a mint which exogenously generates fresh new coins if it is active in period t , $M_n[t] \geq 0$. Assuming that minting is exogenous means that we model the *benefit* of minting (seigniorage) but not its *cost*.³⁰ In the middle of period t , $L_n[t]$ identical workers save a fraction $s_n[t]$ of their coins, and spend the rest on consumption expenditures, $X_n[t]$. Importantly, expenditures contain spending on all goods, possibly including capital goods. What we label saving, $s_n[t]$, solely captures saving into nominal financial assets (coins), not investment into physical capital.

Workers face the following budget constraint,

$$X_n[t] = (1 - s_n[t]) \left((1 - \lambda_n[t]) S_n[t] + M_n[t] \right), \text{ with } s_n[t] \geq 0, \quad (2)$$

where we assume workers cannot borrow ($s_n[t] \geq 0$). In this ‘coin-in-advance’ economy, consumption is financed by currently available coins, and not by promised future income.³¹

At the end of period t , workers earn a competitive wage $w_n[t]$ selling goods in exchange for coins. The stock of coins set aside in location n evolves recursively,

$$S_n[t + 1] = (1 - \lambda_n[t]) S_n[t] + M_n[t] + w_n[t] L_n[t] - X_n[t]. \quad (3)$$

²⁶Earlier writers, including Polanyi (1977), were yet unaware of the vast amount of coins that would be unearthed from the 1980’s onwards through the use of metal detectors.

²⁷We present in Supplemental Appendix C a simple extension of our model with both monetized and barter exchanges, inspired by Kiyotaki and Wright (1989). We show in section 5.5 that our estimates with and without endogenous monetization are highly correlated. As we are not aware of systematic records of the degree of monetization in Late Antiquity, we opted to use a simpler model with no barter as our baseline.

²⁸Gandal and Sussman (1997) discuss medieval weight and fineness measurement technologies.

²⁹We think of λ primarily as coins melted into bullion for their precious metal, possibly as a seigniorage tax to be re-minted. Only a small fraction is literally lost and becomes part of our data.

³⁰We present evidence in section 6.1 that exploitation of ancient mineral deposits is associated with higher minting. This suggests that a lower cost of mining is associated with more minting.

³¹If this assumption were relaxed, promised labor income (in coins) would buy goods in other locations, become income, buy goods again, etc, all within a period. Data on coins would no longer contain information on gross trade.

Trade. Within each period trade is modeled as in [Eaton and Kortum \(2002\)](#). Consumers in location n spend a fraction $\pi_{ni}[t]$ of their expenditures on imports from location i , $X_{ni}[t]$,

$$\pi_{ni}[t] = \frac{X_{ni}[t]}{X_n[t]} = \frac{(w_i[t]/T_i^{1/\theta}[t]d_{ni}[t])^{-\theta}}{\sum_k (w_k[t]/T_k^{1/\theta}[t]d_{nk}[t])^{-\theta}}. \quad (4)$$

Trade shares depends on relative factor prices per efficiency unit (wages, $w_i[t]$, relative to productivity, $T_i^{1/\theta}[t]$), relative bilateral trade costs ($d_{ni}[t]$), and the trade elasticity ($\theta > 0$).

Inter-temporal allocations. Workers have log-utility over real consumption,

$$U_n[t] = \mathbb{E}_t \left[\sum_{\tau \geq t} \beta^{\tau-t} \ln \left(\frac{X_n[\tau]}{p_n[\tau]} \right) \right], \text{ with } p_n[t] = \gamma \left(\sum_k \left(\frac{w_k[t]}{T_k^{1/\theta}[t]} d_{nk}[t] \right)^{-\theta} \right)^{-1/\theta}.$$

β is the discount rate, $p_n[t]$ the ideal price index in n at t , and γ is a constant ([Eaton and Kortum, 2002](#)). Each location chooses a sequence of coins stocks to maximize utility given wages subject to the no borrowing constraint (2), the budget constraint (3), the optimal within-period allocation across imports (4). A transversality condition prevents holding coins forever.

$$\max_{\{S_n[\tau]\}_{\tau \geq t}} \mathbb{E}_t \left[\sum_{\tau \geq t} \beta^{\tau-t} \ln \left(\frac{(1 - \lambda_n[\tau]) S_n[\tau] + M_n[\tau] + w_n[\tau] L_n[\tau] - S_n[\tau + 1]}{\gamma \left(\sum_k (w_k[\tau]/T_k^{1/\theta}[\tau] d_{nk}[\tau])^{-\theta} \right)^{-1/\theta}} \right) \right] \quad (5)$$

$$\text{s.t. } S_n[\tau + 1] \geq w_n[\tau] L_n[\tau], \forall (\tau \geq t), \text{ and } \lim_{\tau \rightarrow \infty} \beta^\tau \frac{S_n[\tau + 1]}{X_n[\tau]} = 0.$$

Saving ($s_n[\tau] \geq 0 \Leftrightarrow S_n[\tau + 1] \geq w_n[\tau] L_n[\tau]$) is used for optimal consumption smoothing.

Dynamic equilibrium. Wages are determined by market clearing, given the trade shares, $\pi_{ni}[t]$ from equation (4), and the coin stock policy function, $S_n[t]$ from equation (5),

$$w_i[t] L_i[t] = \sum_n \pi_{ni}[t] \left((1 - \lambda_n[t]) S_n[t] + M_n[t] + w_n[t] L_n[t] - S_n[t + 1] \right), \forall (i, t). \quad (6)$$

In a useful benchmark where agents save little to none, $s_n[t] \approx 0$, market clearing simplifies to

$$w_i[t] L_i[t] = \sum_n \pi_{ni}[t] \left((1 - \lambda_n[t]) w_n[t - 1] L_n[t - 1] + M_n[t] \right), \forall (i, t). \quad (7)$$

Steady state equilibrium. A steady state where all variables are constant clarifies the mechanics of the model. If agents correctly anticipate they are in a steady state, there is no consumption smoothing motive, so that $s_n = 0$, $X_n = (1 - \lambda_n) w_n L_n + M_n$, and $S_n = w_n L_n$. If agents incorrectly anticipate shocks and save for precautionary motives, the same equality between

expenditure and income holds to a first order, $X_n = \frac{1-s_n}{1-s_n+\lambda_n s_n} ((1-\lambda_n) w_n L_n + M_n)$ with $\frac{1-s_n}{1-s_n+\lambda_n s_n} \approx 1$.³² Equilibrium wages jointly clear markets given the trade equilibrium,

$$w_i L_i = \sum_n \pi_{ni} ((1-\lambda_n) w_n L_n + M_n), \text{ and } \pi_{ni} = \frac{(w_i/T_i^{1/\theta} d_{ni})^{-\theta}}{\sum_k (w_k/T_k^{1/\theta} d_{nk})^{-\theta}}, \quad (8)$$

and the aggregate stock of coins in circulation is constant, $\sum_n M_n = \sum_n \lambda_n w_n L_n$.³³

3.2. Dynamic accumulation in coin hoards

Our aim is to match the quantities in our model to our data on ancient coins which contain, for each coin, information on mint and hoard location and date. To do so, we explicitly follow the movements of individual coins as they travel through the trade network.

We denote by $S_{mi}[t, \tau]$ the number of coins minted in location m at time t which are part of the coin stock of location i at time τ , with $S_i[\tau] = \sum_{m=1}^N \sum_{t < \tau} S_{mi}[t, \tau]$. Coins start their ‘coin life’ when they are minted, so $S_{mm}[t, t] = M_m[t]$. Subsequently, they circulate across locations as they are used for transactions or saved,

$$S_{mi}[t, \tau + 1] = \sum_{n=1}^N (1 - s_n[\tau]) (1 - \lambda_n[\tau]) S_{mn}[t, \tau] \pi_{ni}[\tau] + s_i[\tau] (1 - \lambda_i[\tau]) S_{mi}[t, \tau]. \quad (9)$$

At time τ , each location n has a stock of coins set aside. A fraction $(1 - s_n[\tau])$ is spent on goods (not saved). Of those coins, $(1 - \lambda_n[\tau]) S_{mn}[t, \tau]$ were minted in location m at time t . Consumers in n send a fraction $\pi_{ni}[\tau]$ of those coins to i to pay for imported goods. We assume that coins are fungible so buyers draw coins at random, and $(1 - s_n[\tau]) (1 - \lambda_n[\tau]) S_{mn}[t, \tau] \pi_{ni}[\tau]$ coins minted in location m at time t move from n to i at time τ on average. Summing across all (coin) origins we derive the first term (sum) in equation (9). In addition, a fraction $s_i[\tau]$ of coins is saved locally and remains in region i , the second term in equation (9). We can express the dynamic evolution of the composition of coin stocks in a compact matrix form, $\mathbf{S}[t, \tau] = \mathbf{M}[t]$ and $\mathbf{S}[t, \tau + 1] = \mathbf{S}[t, \tau] (\mathbf{I} - \boldsymbol{\lambda}[\tau]) \tilde{\boldsymbol{\Pi}}[\tau]$, and solve it forward,

$$\mathbf{S}[t, T] = \mathbf{M}[t] \left(\prod_{\tau=t}^{T-1} (\mathbf{I} - \boldsymbol{\lambda}[\tau]) \tilde{\boldsymbol{\Pi}}[\tau] \right) \forall T \geq t, \text{ with } \tilde{\boldsymbol{\Pi}}[\tau] \equiv (\mathbf{I} - \mathbf{s}[\tau]) \boldsymbol{\Pi}[\tau] + \mathbf{s}[\tau]. \quad (10)$$

$\mathbf{S}[t, T]$ is the square $N \times N$ matrix of coin stocks with $(n, i)^{th}$ element $S_{ni}[t, T]$. $\mathbf{M}[t]$ is a diagonal $N \times N$ matrix of minting with n^{th} element $M_n[t]$. $\mathbf{I}$ is the $N \times N$ identity matrix and $\boldsymbol{\lambda}[\tau]$ is a diagonal $N \times N$ matrix of coin loss with n^{th} element $\lambda_n[\tau]$. $\tilde{\boldsymbol{\Pi}}[\tau]$ is the square

³²We estimate $\lambda = 1.7\%$ p.a. (section 4.1); Scheidel (2015) calculates $s = 1.5\%$; so $\lambda s = 0.000255 \approx 0$.

³³If aggregate minting increases (decreases), wages increase (decrease), leading to global inflation (deflation).

Our model is analogous to Dekle, Eaton, and Kortum (2007), with trade deficits and surpluses. Expenditures minus income (the trade deficit of location n , $D_n \equiv X_n - w_n L_n = M_n - \lambda_n w_n L_n$) is equal to the net creation of coins. Any non-mint location runs a trade surplus ($D_n < 0$), and a mint location runs a trade deficit if minting is large enough, with at least one mint location weakly running a trade deficit each period.

$N \times N$ matrix which governs bilateral coin flows. This ‘augmented’ trade matrix $\tilde{\Pi}[\tau]$ is a function of $s[\tau]$, the diagonal $N \times N$ matrix of net saving rates with n^{th} element $s_n[\tau]$, and $\Pi[\tau]$, the $N \times N$ trade matrix with $(n, i)^{th}$ element $\pi_{ni}[\tau]$.

Equation (10) forms the basis of our estimation. Before describing this estimation, we isolate two original features of our model, which help gain intuition on how we can extract information about trade from coins, and clarify the distinctions between data on coins and data on trade.

Coins as a medium of exchange versus a store of value. The stock of coins $\mathbf{S}$ in equation (10) diffuses across locations not according to the trade matrix Π , but to the ‘augmented’ trade matrix $\tilde{\Pi}$. Both matrices have almost the same structure, with one distinction: coins, unlike goods, have an additional tendency to stay locally, because they are also used as a store of value for saving. To make this distinction explicit, we decompose the ‘augmented’ trade share $\tilde{\pi}_{ni}$ into three multiplicative buyer ($\tilde{\alpha}_n$), seller ($\tilde{\beta}_i$), and bilateral ($\tilde{\delta}_{ni}$) terms,

$$\begin{aligned} \tilde{\pi}_{ni}[\tau] &= \tilde{\alpha}_n[\tau] \tilde{\beta}_i[\tau] \tilde{\delta}_{ni}[\tau], \quad \text{with} \\ \tilde{\alpha}_n[\tau] &= 1 / \sum_k \tilde{\beta}_k[\tau] \tilde{\delta}_{nk}[\tau], \quad \tilde{\beta}_i[\tau] = \left(w_i[\tau] / T_i^{1/\theta}[\tau] \right)^{-\theta}, \quad \text{and} \\ \tilde{\delta}_{ni}[\tau] &= \frac{(d_{ni}[\tau])^{-\theta}}{(d_{nn}[\tau])^{-\theta}} \times \begin{cases} 1 & \text{if } n = i, \\ (1 - s_n[\tau] / \tilde{\pi}_{nn}[\tau]) & \text{if } n \neq i. \end{cases} \end{aligned} \quad (11)$$

The classical [Eaton and Kortum \(2002\)](#) trade matrix Π has almost the exact same structure: $\pi_{ni} = \alpha_n \beta_i \delta_{ni}$, where $\alpha_n = 1 / \sum_k \beta_k \delta_{nk}$, $\beta_i = (w_i / T_i^{1/\theta})^{-\theta}$, and $\delta_{ni} = (d_{ni})^{-\theta} / (d_{nn})^{-\theta}$ are buyer, seller, and bilateral terms. In the absence of saving, $s_n = 0$, both matrices are identical. But if $s_n > 0$ the home bias for coins flows is magnified compared to the home bias in trade flows, i.e. the gap between the (high) within-location flows versus the (low) between-location flows is larger for coins than for trade.

Equation (11) shows that coin and trade flows have the same gravity structure up to a single scalar, so that coin flows and the saving rate are sufficient to recover trade flows.

Coins are (re)used for multiple transactions. The second key distinction between coin and trade flows is that coins do not travel just once: they may be used repeatedly throughout their life. This is made explicit by the product of ‘augmented’ trade matrices in equation (10). Our structural estimation unpacks the different elements of the product of those matrices, leveraging the overlapping yet distinct information contained in young versus old coins — which have traveled through a few versus many iterations of the ‘augmented’ trade matrices.

A naive gravity estimation that combines all coins of different ages, wrongly ignoring the dynamic nature of coin flows, would not identify the parameters of the ‘augmented’ trade matrix. This can be most easily seen in a stationary version of our model, though the result extends to non-stationary cases. In a stationary equilibrium there is no net saving, $s[\tau] = 0, \forall \tau$, and the trade matrix and ‘augmented’ trade matrix are identical and time invariant, $\tilde{\Pi} = \Pi$. The dynamics of coin stocks in equation (10) simplify into

$$\mathbf{S}[t, t+a] = \mathbf{S}[a] = \mathbf{M} \left((\mathbf{I} - \boldsymbol{\lambda}) \Pi \right)^a, \forall (t, a). \quad (12)$$

Only age (a) matters, and not calendar time (t). Combining coins of different ages, we get

$$\sum_{a=0}^A \mathbf{S}[a] = \mathbf{M} \left(\sum_{a=0}^A \left((\mathbf{I} - \boldsymbol{\lambda}) \Pi \right)^a \right)_{A \rightarrow +\infty} = \mathbf{M} \left(\mathbf{I} - (\mathbf{I} - \boldsymbol{\lambda}) \Pi \right)^{-1}. \quad (13)$$

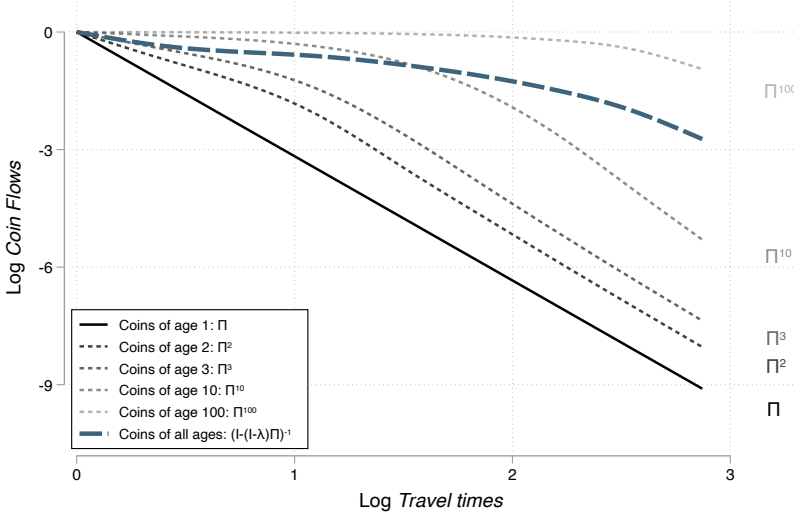


FIGURE 5.—Flows of coins of different ages. *Notes:* This figure presents a numerical illustration of the flow of coins of different ages as a function of trade costs. We use equations (12) and (13), and the trade model (4) to simulate an economy with 40 locations around a regular polygon, with same technology $T_n = T, \forall n$, a trade elasticity $\theta = 4$, and an annual coin loss rate $\lambda = 0.017$. Locations are symmetric so wages are equalized and trade shares simplify to $\pi_{ni} = (d_{ni})^{-\theta} / \sum_k (d_{nk})^{-\theta}$. We parameterize $(d_{ni})^{-\theta} = (\text{TravelTimes}_{ni})^{-\zeta}$ with $\zeta = 3.17$ —see our structural estimation in table III. Log travel times are on the x-axis; and log flows of coins of different ages on the y-axis — $\ln(S_{ni}[t, t+a])$ for age a . To ease comparisons, we normalize the largest log flows and smallest log travel times to zero, so all curves start at coordinate (0,0). ‘Naively’ combining coins of all ages, i.e. treating the Leontief inverse $(I - (I - \lambda)\Pi)^{-1}$ — thick dashed line — as if it were the trade matrix Π — solid line, gives a misspecified travel times elasticity of 1.07 (OLS estimate), substantially below the true $\zeta = 3.17$.

The share of coins from different mint origins in different destinations depends not on the trade matrix Π , but on the Leontief inverse of the trade matrix discounted by $(I - \lambda)$: $(I - (I - \lambda)\Pi)^{-1}$. The reason is simple: newly minted coins percolate through the trade network, just as value added shocks percolate through the input-output network in the work of Wassily Leontief (Leontief, 1941, 1944). The intensity with which coins flow from one location to another depends on bilateral trade shares, just as the intensity with which one upstream sector affects the production of a downstream sectors depends on bilateral input shares. The same coin will travel multiple times through the trade network (until hit by a Poisson death shock λ), just as value added travels multiple times through the input-output network. However, unlike in conventional static models of input-output linkages, coins take time to percolate through the system, similar to the dynamic model of Liu and Tsyvinski (2024).

Figure 5 illustrates the potential bias from wrongly interpreting coin flows as trade flows and ignoring coin ages. Within the first period of their life, coin flows mirror trade flows (Π in figure 5). The same elasticity governs coin and trade flows, $S_{ni}[t, t+1] \propto \pi_{ni} \propto (d_{ni})^{-\theta}$. In the second period of their life, coins have traveled twice through the trade network (Π^2 in figure 5). The negative impact of trade costs over short distances weakens as coins have started diffusing within nearby destinations. As coins age, their flows gradually escape the negative effect of trade costs; coins diffuse through the trade network and converge towards a uniform

distribution (see the flattening slopes of $\Pi, \Pi^2, \dots, \Pi^{100}$ in figure 5).³⁴ A naive estimation using coins of all ages combined, i.e. wrongly interpreting the Leontief inverse $(\mathbf{I} - (\mathbf{I} - \lambda)\Pi)^{-1}$ as if it were the trade matrix Π , would infer incorrect parameters. In our numerical example, if we assume that trade depends on travel times, $(d_{ni})^{-\theta} = (\text{TravelTimes}_{ni})^{-\zeta}$, with a true elasticity $\zeta = 3.17$, we would wrongly estimate an elasticity of 1.07. This corresponds approximately to the discrepancy between our reduced-form estimate combining coins of all ages (1.14 in table I) and our upcoming structural estimate (3.17 in column 2 of table III).³⁵ Supplemental Appendix B.5 and figure B.4 show reduced-form evidence suggestive of this phenomenon in our coin dataset: across separate gravity regressions (as in table I) for coins of different ages, the elasticity in absolute value falls towards zero as we move from younger to older coins.

4. ESTIMATION

4.1. Mapping the model to the data

To map the model to the data, we need to define the empirical counterparts to time periods, t , locations, n , the coin loss rate, λ , and we parameterize trade costs as a function of observables.

Definition of time periods. We aggregate mint and hoard dates (tpq) to 20-year intervals, and show robustness for shorter (10-year) and longer (30-year) intervals.

Definition of locations. We partition the world into $N = 13$ regions (see figure 6). In the Islamic world they correspond to aggregates of provinces of the Umayyad caliphate (Cornu, 1983). In the Roman world they are based partly on Roman provincial borders, and partly on 9th-century political borders (see Supplemental Appendix A for additional details).

Assumption: constant loss rate. With a constant loss rate λ , any collection of coins minted at time t will gradually disappear from the monetary system. At time $t + 1$ only a fraction $(1 - \lambda)$ remains, at time $t + 2$ a fraction $(1 - \lambda)^2$, etc. The same exponential decay holds for any starting date t , and it holds for the random sample found by archaeologists. We aggregate all the coins in our dataset, and express the density of coins of age a as $f(a) \propto (1 - \lambda)^a$, corresponding to figure 3b. Taking logs, we estimate $\ln(1 - \lambda)$ by OLS,

$$\ln f(a) = \text{constant} + \ln(1 - \lambda) \times a + \varepsilon(a). \quad (14)$$

We estimate a coin loss rate of 1.7% per year, or 30% over 20 years, $\hat{\lambda}_{20\text{-year}} = 0.3$.³⁶

Parameterization of trade costs. We assume that bilateral trade costs scaled by the trade elasticity θ are a function of (directed) bilateral travel times, and a possible ad-valorem proportional

³⁴Our model of gradual diffusion of coins through a trade network is the discrete time and space analog to the continuous time and space model of diffusion of information through a trade network in Chaney (2018).

³⁵Figure 5 is a stylized example with symmetric locations around a regular polygon, meant to build intuition, not to resemble the real ancient world. The proximity between 1.07 in figure 5 and 1.14 in table I is coincidental.

³⁶See Supplemental Appendix B.6 and table B.V for formal estimation results. Our estimate of an annual coin loss rate of 1.7% is remarkably similar to that estimated by numismatists, who use radically different methods. For Roman times, the most frequently cited estimate is by Crawford and Hopkins, of 2% per annum (Hopkins, 1980). Patterson (1972) reports an estimate of 2% for Roman times (AD 150), and 1% for Islamic times (AD 800). These numbers are based on estimates of mint output from information on the number of dies used to mint coins, and multiplying die counts by estimates of the average number of coins that were struck with one die. The comprehensive die studies that are required for this procedure are, however, only available for a small subset of mints and time periods.

penalty incurred when crossing political or religious borders,

$$\ln \left((d_{ni}[t])^{-\theta} \right) = \kappa_0 - \zeta \ln (\text{TravelTime}_{ni}) \quad (15)$$

$$- \kappa_{\text{Pol.}} \text{PoliticalBorder}_{ni}[t] - \kappa_{\text{Rel.}} \cdot \text{ReligiousBorder}_{ni}[t], \forall (n \neq i, t).$$

We normalize $d_{nn}[t] = 1, \forall n, t$, as in [Eaton and Kortum \(2002\)](#). The constant κ_0 adjusts travel time units, and governs the home bias in trade. $\kappa_{\text{Rel.}}$ is either a scalar or a vector distinguishing the eastern, western, and Mediterranean borders, $\kappa_{\text{Rel.}} = (\kappa_{\text{Rel.}}^{\text{East}}, \kappa_{\text{Rel.}}^{\text{West}}, \kappa_{\text{Rel.}}^{\text{Med.}})$ in our baseline.³⁷ From our ‘augmented’ trade model, which describes coin flows driven both by trade and saving, we derive the bilateral determinants of coin flows, $\tilde{\delta}_{ni}[t]$ in equation (11),

$$\ln \left(\tilde{\delta}_{ni}[t] \right) = \tilde{\kappa}_0 - \zeta \ln (\text{TravelTime}_{ni}) \quad (16)$$

$$- \kappa_{\text{Pol.}} \text{PoliticalBorder}_{ni}[t] - \kappa_{\text{Rel.}} \cdot \text{ReligiousBorder}_{ni}[t], \forall (n \neq i, t),$$

and $\tilde{\delta}_{nn}[t] = 1, \forall (n, t)$. The bilateral determinants of external trade flows, $(d_{ni}[t])^{-\theta}$, and coin flows, $\tilde{\delta}_{ni}[t]$, only differ by a multiplicative scalar, $e^{\tilde{\kappa}_0 - \kappa_0}$, due to saving. Given within region coin flows, $\tilde{\pi}_{nn}[t]$, this scalar maps into the saving rates,

$$s_n[t] = \tilde{\pi}_{nn}[t] (1 - e^{\tilde{\kappa}_0 - \kappa_0}). \quad (17)$$

$\tilde{\pi}_{nn}[t]$ controls the home bias in coins, and $(1 - e^{\tilde{\kappa}_0 - \kappa_0})$ adjusts for the discrepancy (due to saving $s_n[t]$) between the home bias in coins (governed by $\tilde{\kappa}_0$) and the home bias in trade (governed by κ_0). In the absence of direct evidence on ancient trade, we cannot directly estimate κ_0 . Instead, we choose κ_0 to match an average ancient saving rate into nominal assets of 1.5% ([Scheidel, 2020](#)). This estimated savings rate for Roman times is likely an upper bound for Late Antiquity, where property rights were weaker and conflict more widespread.

Travel times. To compute (optimal) travel times given the transportation network and technology we use two geo-spatial models constructed by historians to provide quantitative estimates of (shortest) trade routes and trade costs. The first is Orbis ([Scheidel, 2015](#)), a directed graph of cities and trade routes of the Roman world (i.e. from Britannia in the north-west to Egypt, Palestine, and Syria in the south-east) along with a calibrated model of trade costs, in monetary units and units of time, along the edges to allow for the calculation of shortest paths. The second is al-Turayyā ([Romanov and Seydi, 2022](#)), a digitalization of the Atlas of the Islamic World of [Cornu \(1983\)](#), which, similarly, contains the coordinates of cities and trading posts connected by trade routes, but without estimates of travel times. We combine the nodes of al-Turayyā and Orbis and extend [Scheidel \(2015\)](#)’s methodology from Orbis to calculate travel times for the Islamic world (see Supplemental Appendix A for details).

Figure 6 shows the combined graph. We validate the resulting travel times by comparing them to those reported by the 10th-century Arab geographer [Al-Muqaddasī \(985\)](#) (see Supplemental Appendix B.7 and figure B.5). For each region, we project the weighted average of mint locations (with the shares of each location in total coin output as weights) to the closest

³⁷Our parameterization for trade costs is potentially inconsistent with the assumption of arbitrage trade because we compute optimal travel routes once, without taking into account the additional costs associated with potentially multiple border crossings. In practice, this does not happen: we manually verify that the estimated costs in equation (15) cannot be lowered by taking a longer route avoiding unnecessary border crossings.

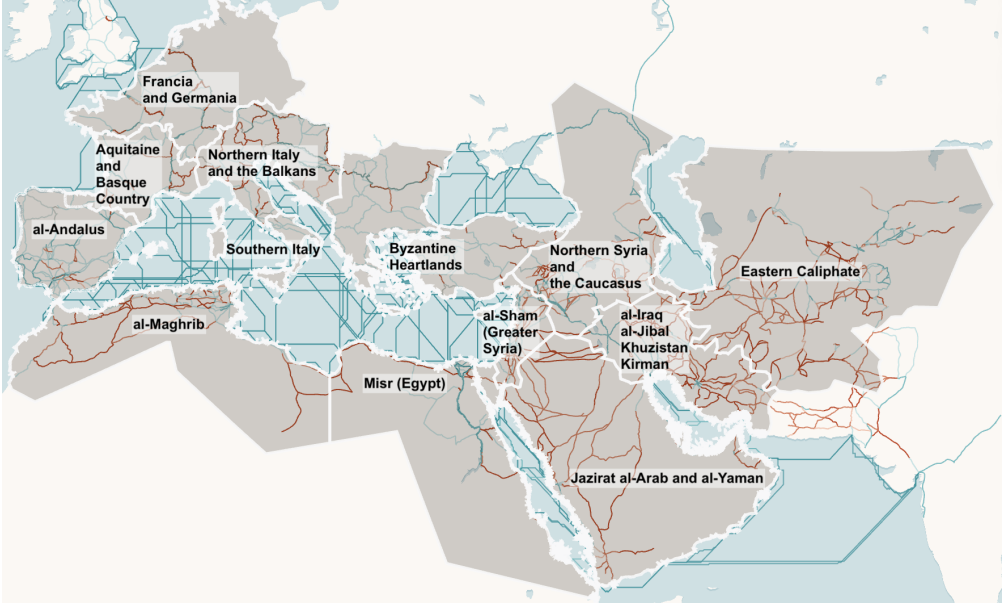


FIGURE 6.—The combined geospatial model. *Notes:* The figure shows the combined geospatial models from Orbis and al-Ṭurayyā, along with our thirteen regions. Edges in blue (red) indicate faster (slower) travel speeds.

vertex on the graph. The shortest travel time between the central vertices in n and i is our time-invariant measure of $TravelTime_{ni}$.

Political and religious borders. We construct political border and religious border dummies by coding the start and end years of the presence of political entities across regions. We set $PoliticalBorder_{ni}[t]$ to one if the set of political entities that occupy at least some part of the regions n and i for some part of the 20-year time interval is disjoint. $ReligiousBorder_{ni}[t]$ captures the border between the emerging religion of Islam and the rest of the world. We use two alternative specifications. The first specification is a dummy equal to one if all political entities in region n are Islamic and none in i are, or vice versa. The second specification — our baseline — distinguishes the eastern, western, and Mediterranean borders of Islam. The eastern land border is between the Byzantine heartlands and the Caliphate regions east of Egypt, the western land border is between al-Andalus and Aquitaine or Francia/Germania, and the Mediterranean maritime border is between all other region pairs.

Coin hoard data generating process. We assume that our hoard dataset $\mathbf{H}$ is a random sample from the stocks of coins in circulation in each region and period. We group all hoards within a region and period, with $H_h[T]$ the total amount of coins found in region h and buried at time T (tpq), which we decompose into coin types, with $H_{m,h}[t, T]$ the amount of coins minted in m at time t within that hoard. Our random sampling assumption means that the expected share of coin types within a hoard equals the share of coin types within a coin stock in our model (10),

$$\mathbb{E} \left[\frac{H_{m,h}[t, T]}{H_h[T]} \right] = \frac{S_{m,h}[t, T]}{S_h[T]}. \quad (18)$$

As we discuss in section 2, we recognize that the probability that a coin ends up in our dataset may vary between regions and periods, depending on whether coins were lost and deposited in

the ground, found by archaeologists, and documented by experts. By using only information on the shares of coins *within* hoards, we condition on those events being realized (lost, found, documented), and we purge any variation in the probability of those events.³⁸

Supplemental Appendix B.8 presents suggestive evidence in support of our random sampling assumption. For eight region $\times$ periods with the largest number of coins and hoards, we randomly split hoards in two subsets, and confirm that the geographic (domestic vs foreign) and age distributions of coins in each subsample of hoards within a region $\times$ period are statistically indistinguishable (see table B.VI). We further show that for fifty one out of fifty six pairs of region $\times$ periods, the geographic (domestic vs foreign) and age distributions of coins in both region $\times$ periods are statistically different from each other (see table B.VII). This suggests that the geography and the age distribution of coins are similar within a region $\times$ period unit but distinct between units, a necessary condition for random sampling.

Coin accounting. We use two different accounting methods for the amounts of coins within hoards, $H_{m,h}[t, T]$: either the count of coins combining all denominations together, or the value of gold and silver coins accounting for weight, denomination, and metal content as in section 2 (see footnote 22 for details on computing relative gold and silver values), discarding undetermined coins and bronze coins for which relative values fluctuate too much and which are poorly documented. Each method has advantages and drawbacks. Using counts minimizes the amount of data we discard, but ignores relative values. Using values is closer to our assumption that coins are fungible, but forces us to discard information on 75% of the coins in our dataset. We use counts as our baseline, and present robustness using values.

4.2. Estimation method

We estimate the structural parameters of our model by maximum likelihood. Given our random sampling assumption (18), the probability of observing $(\dots, H_{m,h}[t, T], \dots)_{m,t \leq T}$ coins minted in different regions (m 's) at different times (t 's) among the total of $H_h[T]$ coins within a hoard buried in region h at time (tpq) T is multinomial, and a function of coin stocks,

$$\Pr(\dots, H_{m,h}[t, T], \dots) = \frac{H_h[T]!}{\prod_{m', t' \leq T} H_{m',h}[t', T]!} \prod_{m, t \leq T} \left(\frac{S_{m,h}[t, T]}{S_h[T]} \right)^{H_{m,h}[t, T]}.$$

It depends on the model parameters through the predicted coin shares, $S_{m,h}[t, T]/S_h[T]$, governed by the coin flow recursive equation (10), the coin flow static equation (11), which depend on the coin loss rate (14), and the bilateral determinants of coin flows (16). We collect all parameters in the vector Θ : the saving and trade costs parameters, $\tilde{\kappa}_0$, ζ , $\kappa_{\text{Pol.}}$, and $\kappa_{\text{Rel.}}$ from (16), time-varying minting $M_n[t]$ from (10), and time-varying seller terms $\tilde{\beta}_n[t]$ from (11),

$$\Theta = \left(\tilde{\kappa}_0, \zeta, \kappa_{\text{Pol.}}, \kappa_{\text{Rel.}}, (\dots, M_n[t], \dots)_{n \neq n_0, t \neq t_0}, (\dots, \tilde{\beta}_n[t], \dots)_{n \neq n_0, t}, \right).$$

As we target coin shares within hoards, we can never recover the total number of coins minted over 320-950. We normalize mint output $M_{n_0}[t_0] = 100$ for an arbitrary region n_0 (northern

³⁸Our assumption of random sampling would be violated if, for example, hoards with certain types of coins were more likely to be described in the literature. This concern is mitigated by the fact that many of the regions we study are covered by catalogues of coin finds that seek to exhaustively describe all finds on a territory.

Italy) and period t_0 (320-340). Similarly, only *relative* seller terms matter for coin flow shares within each period, so we normalize $\tilde{\beta}_{n_0}[t] = 100$ for region n_0 in each period t .

To estimate Θ , we maximize the log-likelihood of observing our coin hoards sample,

$$\hat{\Theta} = \arg \max_{\Theta} \sum_{h,T} \sum_{m,t \leq T} H_{m,h}[t,T] \left(\ln S_{mh}[t,T](\Theta) - \ln \sum_{m',t' \leq T} S_{m'h}[t',T](\Theta) \right). \quad (19)$$

Given those structural estimates, we recover the parameter κ_0 governing bilateral trade costs — distinct from the parameter $\tilde{\kappa}_0$ governing bilateral coin flows in the presence of saving — using equation (17) and an average net ancient saving rate into coins of 1.5% (Scheidel, 2020),

$$\kappa_0 \text{ s.t. } (1 - e^{\tilde{\kappa}_0 - \kappa_0}) \mathbb{E}_{n,t} [\tilde{\pi}_{nn}[t]] = 0.015. \quad (20)$$

With those estimates, we can recover all equilibrium variables (see section 5.1).

Identification. We present an asymptotic identification proof and explore identification in finite sample using numerical simulations in Supplemental Appendix E. The identification of the trade costs parameters is relatively straightforward: by combining at least two consecutive series of coin hoards, we can isolate both minting and buyer terms into coin origin fixed effects, seller terms into coin destination fixed effects, and estimate the bilateral determinants of coin flows. Intuitively, the relative frequency of coins of different origins across hoards directly informs the bilateral component of coin flows. Separately identifying minting from seller terms is harder, both because by using only coin shares within hoards we do not have direct information on the relative scale of minting from different regions and periods, and because minting affects expenditure in one period but becomes income in the next and therefore gets jumbled up with production, governed by the seller terms. Combining information from hoards in more consecutive periods than regions and the structure of the model allows to break this indeterminacy. A further complication arises with sparse data, where many observations are zeros. Numerical simulations suggest that our estimator recovers known parameters accurately with sparse data even if the number and sizes of hoards are systematically biased — because we use shares of coins within hoards — and that misspecification biases may exist but are unlikely to be large.

4.3. Parameter estimates

Ancient trade costs. Table III shows the estimates of the parameters governing ancient trade costs. Column 2, which distinguishes the religious border in the east (in and out of Byzantium), in the west (in and out of al-Andalus), or over the Mediterranean in and out of its non-Islamic northern shore, is our baseline. We discuss the robustness of our estimates to alternative coin accounting methods and time aggregations in section 5.5.

In our simpler specification (column 1), the travel time elasticity of trade, $\zeta = 3.21$ (s.e. 0.02), is close to the 2.05-2.89 range of estimates from FHLL+ (2022) using bilateral trade in terra sigillata in ancient Rome and optimal travel times along the Roman transportation network, and to the 1.9 distance elasticity from Barjamovic, Chaney, Coşar, and Hortaçsu (2019) using merchant records in Bronze Age Anatolia. This proximity to estimates using actual — though partial — ancient trade data is reassuring, as we do not use any direct information on trade, but only indirect information from coins. This travel time elasticity is robust to alternative specifications for the religion border effect, 3.21 versus 3.17 in column 1 versus 2.

In our simpler specification (column 1), the political ($\kappa_{\text{Pol.}}$) and religious ($\kappa_{\text{Rel.}}$) border effects are large, but of a magnitude similar to estimates for modern border effects. Assuming a

TABLE III
BILATERAL DETERMINANTS OF ANCIENT TRADE

	Log Trade	
	(1)	(2)
Log Travel Time	-3.21 (0.02)	-3.17 (0.02)
Political Border	-0.27 (0.02)	-0.54 (0.02)
Religious Border	-4.03 (0.11)	
Religious Border: East		-4.28 (0.22)
Religious Border: West		-1.84 (0.12)
Religious Border: Mediterranean		-5.0 (0.19)
Sample	All	All
Coin Accounting	Number	Number
Observations	4,403	4,403

Note: The table shows estimates for the trade costs parameters in equation (15). “Political Border” is one if the sets of political entities that occupy at least some part of the regions during the 20-year time period are completely disjoint, and zero otherwise. In column 1 “Religious Border” is one if all political entities in one region are Islamic and all are non-Islamic in the other region, and zero otherwise. In column 2, our baseline, “Religious Border: East” is one iff the religious border dummy is one and the regions are the Byzantine Heartlands and one of the Caliphate regions east of Egypt, or vice versa; “Religious Border: West” is one iff the religious border dummy is one and the regions are al-Andalus and Aquitaine or Francia/Germania, or vice versa; “Religious Border: Mediterranean” is one for all other region pairs where the religious border dummy is one. “Observations” denotes the number of observations (m, h, t, T) in the log-likelihood equation (19) where $H_{m,h}[t, T] > 0$. See Supplemental Appendix table B.VIII for the sensitivity of our estimates to alternative coin accounting methods and definitions of period lengths. Asymptotic standard errors in parentheses.

trade elasticity $\theta = 4$ (Simonovska and Waugh, 2014), those correspond to an 7% ad-valorem tax for crossing a political border ($d_{\text{across}}/d_{\text{within}} = e^{\kappa_{\text{Pol.}}/\theta} \approx 1.07$), and a 174% tax for crossing a religious border ($d_{\text{across}}/d_{\text{within}} = e^{\kappa_{\text{Rel.}}/\theta} \approx 2.74$), of the same magnitude as the estimated 49% cost of crossing the US-Canada border in 1993 (Anderson and Wincoop, 2003).³⁹

Distinguishing the eastern and western land borders from the Mediterranean maritime border (column 2), the political border effect remains small (0.27 versus 0.54 in column 1 versus 2). It does however reveal different estimated penalties associated with crossing from Islamic to non-Islamic regions. The religious border effect is strongest for crossing the Mediterranean ($\kappa_{\text{Rel.}}^{\text{Med.}} = 5.0$) and for the eastern border into Byzantium ($\kappa_{\text{Rel.}}^{\text{East}} = 4.28$), and lowest for the western border from al-Andalus ($\kappa_{\text{Rel.}}^{\text{West}} = 1.84$).

Our structural estimation confirms the reduced form evidence in section 2. The border in and out of Islamic regions becomes costlier to cross *after* the birth of Islam.

Minting output. Figure 7 shows estimates of mint output by region and time interval. Our estimates line up with several patterns described in the numismatic literature: (i) the decline of mint output in the western Mediterranean following the demise of the West Roman Empire in the late 5th century; (ii) the large decline of Byzantine mint output in the ‘Byzantine dark ages’ of the 8th century; (iii) the gradual increase in Arab and Frankish mint output starting from the late 7th century. Importantly, we are not merely counting coins; we *estimate* minting

³⁹ Anderson and Wincoop (2003) estimate that trade is $\exp(1.59) \approx 5$ times larger within than between the US and Canada in 1993. For $\theta = 4$ it corresponds to $d_{\text{across}}/d_{\text{within}} = e^{1.59/4} \approx 1.49$, a 49% ad-valorem border tax.

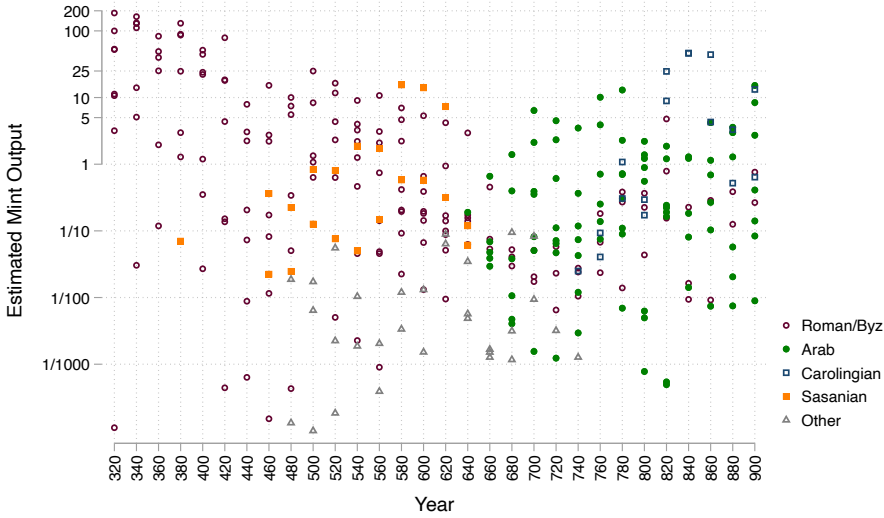


FIGURE 7.—Estimated mint output, AD 320-900. *Notes:* The figure shows the estimated coin output $M_n[t]$ by time t (horizontal axis) and region n , using our baseline specification in column 2 of table III, and broken down by the political entities that locations are—at that time—primarily associated with. The units are relative to northern Italy in 320-340, which we normalize to have a mint output of 100.

output from the relative shares of coins from different mints across hoards, absorbing variation due for instance to differential hoarding patterns or hoard finding rates across space and time.

Seller terms. The last set of estimates, the seller terms, contain combined information about factor prices and technology, $\tilde{\beta}_n[t] = (w_n[t]/T_n^{1/\theta}[t])^{-\theta}$. Those terms are not readily interpretable because factor prices ($w_n[t]$) are endogenous and depend on all other parameters. The structure of our dynamic model of trade and coins allows us to separately recover all equilibrium variables from our parameter estimates, including relative prices, which we turn to next.

5. TRADE AND THE END OF ANTIQUITY

In this section, we first explain how we can recover trade and real consumption from our parameter estimates. We then present granular evidence on the dynamics of ancient trade and consumption. While those findings are mostly descriptive, they demonstrate how using ancient coins, and more broadly a quantitative dynamic model of trade, allows to reveal patterns of a changing economic geography; they also inform and clarify active debates in history.

5.1. Recovering trade, nominal income, and real consumption

Even though we only have data on coin hoards, they implicitly contain information about production, trade, and endogenous prices. From those data we can recover all economic variables using solely our parameter estimates, the structure of our model, and a single unknown parameter, the trade elasticity, arbitrarily set to $\theta = 4$ (Simonovska and Waugh, 2014).⁴⁰

⁴⁰This unknown elasticity θ has no impact on our estimation strategy, which is identical for any choice of θ . It also has no impact on the computation of trade flows (section 5.2) and nominal incomes (section 5.3). It is solely required

Given the low savings rate (1.5% p.a. computed by [Scheidel, 2020](#)), we approximate it to zero to simplify the algebra, but note that our method extends to positive saving with nearly identical results. The inherent sparsity of our ancient coin hoard data introduces noise in our estimates for seller terms and minting. To smooth out some of this noise, we use a simple moving average: for $t \in [380, 880]$, we use $\frac{1}{3} \sum_{\tau=t-1}^{t+1} \tilde{\beta}_n[\tau]$ instead of $\tilde{\beta}_n[t]$, and $\frac{1}{3} \sum_{\tau=t-1}^{t+1} M_n[\tau]$ instead of $M_n[t]$. We do not smooth our time-invariant parameters for bilateral trade costs.

Trade. We recover trade shares using (11) from seller terms, $\tilde{\beta}_i[t]$, and the determinants of bilateral trade, $(\kappa_0, \kappa_{\text{Pol.}}, \kappa_{\text{Rel.}})$ and $(\text{PoliticalBorder}_{ni}[t], \text{ReligiousBorder}_{ni}[t])$ from which we compute $d_{ni}^{-\theta}[t]$ using (15),

$$\pi_{ni}[t] = \frac{\tilde{\beta}_i[t] d_{ni}^{-\theta}[t]}{\sum_k \tilde{\beta}_k[t] d_{nk}^{-\theta}[t]}, \forall (n, i, t). \quad (21)$$

Estimated trade shares are invariant to any normalization of $\tilde{\beta}_i[t]$ and $M_n[t]$, and hold for any θ .

Nominal income. Market clearing imposes a relationship between nominal income in one region, income and minting in other regions, and trade shares. Information on estimated trade shares $\pi_{ni}[t]$, minting $M_n[t]$, and the coin loss rate λ , is sufficient to recover incomes. Markets clear dynamically so we need to make an assumption for period $t_0 - 1$ *before* our sample starts. Absent any data, we simply assume that aggregate incomes in $t_0 - 1$ are the same as in t_0 and use the steady state market clearing condition (8). For every subsequent periods we recover incomes by recursively solving the dynamic market clearing conditions (7).

$$\text{For } t = t_0 : (w_i[t] L_i[t]) = \sum_n \pi_{ni}[t_0] \left((1 - \lambda) (w_n[t] L_n[t]) + M_n[t_0] \right), \forall i.$$

$$\text{For } t > t_0 : (w_i[t] L_i[t]) = \sum_n \pi_{ni}[t] \left((1 - \lambda) (w_n[t-1] L_n[t-1]) + M_n[t] \right), \forall i. \quad (22)$$

Estimated nominal incomes inherit our (single) minting normalization and hold for any θ .

Real consumption. Nominal consumption, $X_n[t] = (1 - \lambda) w_n[t-1] L_n[t-1] + M_n[t]$, normalized by the price index, $p_n[t] = \gamma (\sum_k (w_k[t] / T_k^{1/\theta}[t] d_{nk}[t])^{-\theta})^{-1/\theta}$ with γ a constant, defines real consumption. Simple manipulations yield,

$$\frac{X_n[t]}{p_n[t]} = \gamma^{-1} (\pi_{nn}[t])^{-1/\theta} (T_n^{1/\theta}[t] L_n[t]) \left(\frac{(1 - \lambda) w_n[t-1] L_n[t-1] + M_n[t]}{w_n[t] L_n[t]} \right). \quad (23)$$

The first and last terms are computed from trade shares, nominal incomes, minting, the coin loss rate, and θ . Using $\tilde{\beta}_n[t] = (w_n[t] / T_n^{1/\theta}[t])^{-\theta}$ in the trade equation (11), the middle term is recovered from nominal incomes, seller terms, and θ : $(T_n^{1/\theta}[t] L_n[t]) = (w_n[t] L_n[t]) (\tilde{\beta}_n[t])^{1/\theta}$.

The intuition for the components of real consumption in (23) is as follows. A higher production capacity ($T_n^{1/\theta} L_n$) allows to produce and consume more. As in [Eaton and Kortum \(2002\)](#), trade openness ($\pi_{nn}^{-1/\theta}$) further increases real consumption, leveraging the gains from

to distinguish nominal from real consumption (section 5.4). In practice however, even tripling θ from 4 to 12 has a barely noticeable impact on estimated consumption (see table IV in section 5.5).

trade. As in [Dekle, Eaton, and Kortum \(2007\)](#), trade deficits (surpluses) financed by positive (negative) net coin creation allow a region to consume more (less) than it produces, $X_n[t] = (1 - \lambda)w_n[t - 1]L_n[t - 1] + M_n[t]$ larger (smaller) than $w_nL_n[t]$.

Our normalizations for the seller terms and for minting do not affect the first and last terms, both unit-free ratios, but they affect the middle term. We choose units for technology such that $\sum_n X_n[t]/p_n[t] \equiv 1, \forall t$, so regional consumption within each period is always defined as a share of aggregate consumption. This normalization is an important limitation of our structural estimation: data on nominal trade flows can reveal *relative* aggregate production and consumption, but can never reveal *absolute* levels. This means that our estimates cannot identify any one-period global shock affecting all regions from the Indus to the Atlantic.

5.2. *The geography of Mediterranean trade and the Pirenne thesis*

Our granular estimates for trade and income allows us to paint a detailed picture of the evolution of Mediterranean trade. We partition the Mediterranean into southern—al-Andalus (Iberia), al-Sham (Greater Syria), Misr (Egypt), al-Maghrib—and northern regions—Aquitaine and Basque Country, Northern Italy, Southern Italy, Byzantine Heartlands—and decompose Mediterranean trade to and from northern and southern regions. Figure 8 shows the geographic evolution of Mediterranean trade,⁴¹ partitioned into north-south crossings (panel 8a), south-south crossings (panel 8b), and north-north crossings (panel 8c).

We confirm in panel 8a the hypothesis formulated by [Pirenne \(1939\)](#) that north-south Mediterranean trade falls sharply after the Arab conquests of the Near East, Northern Africa, and the Iberian peninsula. North-south crossings collapse from around 4% of the total income of Mediterranean regions until the late 6th century to below 1% after the late 7th century. This is to be expected given our estimate of a large Mediterranean border effect discouraging trade in and out of Islamic regions. This structural evidence also confirms the reduced form evidence on coins crossing the Mediterranean in section 2.3 (see Supplemental Appendix figure B.3).

However, our estimates also reveal that while north-south crossings fall sharply after the Arab conquests, overall Mediterranean trade remains vibrant throughout Late Antiquity, as south-south crossings compensate the collapse of north-south trade. Panel 8b shows that south-south crossings represents between 5% and 10% of aggregate Mediterranean income, both before and after the Arab conquests. Following the rapid economic expansion of Islamic regions in the early 8th century, south-south crossings actually increase after the Arab conquests, although our estimates are less precise. Panel 8c shows that north-north crossings on the other hand remain small—around 1% of Mediterranean income both before and after the Arab conquests.

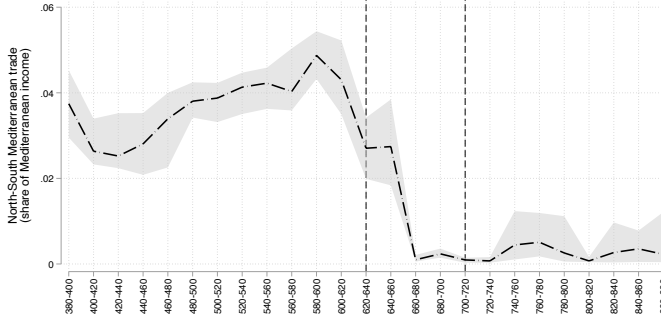
Taken together, our evidence that north-south Mediterranean trade collapses in the 7th century after the Arab conquests, while south-south trade remains stable and even rises, offers both support and nuance to the [Pirenne \(1939\)](#) thesis and to our descriptive evidence.

5.3. *The economic decline of the Mediterranean*

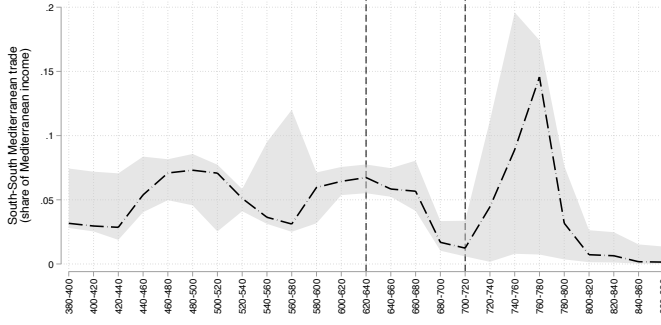
The collapse of north-south Mediterranean trade is mirrored in the overall economic decline of Mediterranean regions. Figure 9a displays time series estimates of the average size of Mediterranean regions relative to the average size of all regions in our sample.⁴² It shows a

⁴¹We define any predicted trade flow from one Mediterranean region to another as Mediterranean, but exclude trade flows within regions. This means, for instance, that we count a shipment from Northern Italy to al-Andalus as Mediterranean, even though it may travel overland through the Alps and the Pyrenees; and we do not count a shipment within Northern Italy as Mediterranean, even though it may travel by sea.

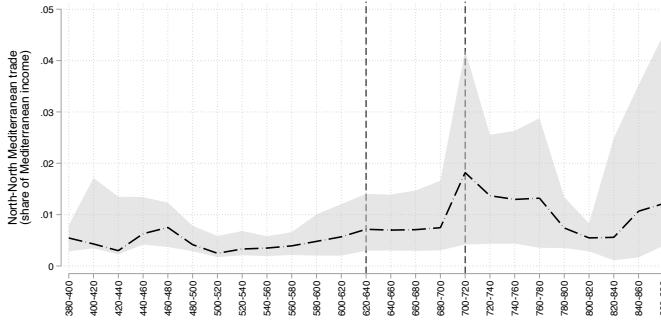
⁴²We use nominal incomes to compute relative sizes. Since those estimated income series contain endogenous relative prices, relative nominal incomes are equivalent to relative real incomes using a single global price index.



(a) North-south Mediterranean trade



(b) South-south Mediterranean trade



(c) North-north Mediterranean trade

FIGURE 8.—The geography of Mediterranean trade, AD 380-880. *Notes:* We partition the Mediterranean into southern—al-Andalus (Iberia), al-Sham (Greater Syria), Misr (Egypt), al-Maghrib—and northern regions—Aquitaine and Basque Country, Northern Italy, Southern Italy, Byzantine Heartlands. We compute exports from origin i to destination n as the product of trade share and importer's income, $\pi_{ni}[t] \times w_n[t]L_n[t]$. Mediterranean trade is partitioned into north-south, south-south, and north-north flows. Each panel shows the time series of one component of Mediterranean trade as a share of total Mediterranean income, $\sum_{n \in \text{Med.}} w_n[t]L_n[t]$: north-south for panel (a), south-south for panel (b), and north-north for panel (c). 95% confidence intervals (shaded areas) are computed from re-estimating our model on 100 bootstrapped samples from our coin hoard data. The vertical lines correspond to the Arab conquests, starting in 622 under Muhammad and ending in 713 with the conquest of Iberia.

gradual decline starting in the early 6th century and continuing at least until the late 9th century. Over four centuries, Mediterranean regions went from being 50% larger to 50% smaller

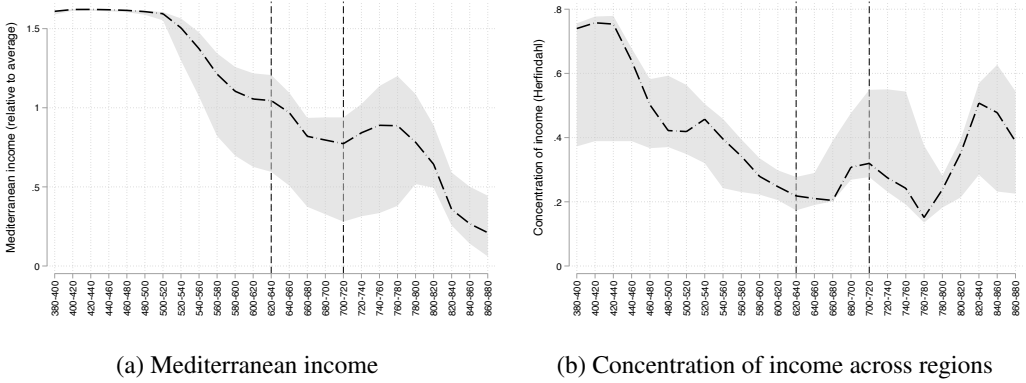


FIGURE 9.—Mediterranean decline and falling economic concentration, AD 380-880. *Notes:* Panel (a) presents the time series of average nominal incomes of the Mediterranean regions relative to the average nominal income of all regions in our sample. Panel (b) presents the time series of Herfindahl indices of income concentration, defined for each period as the sum across all thirteen regions of the squared share of regional income over aggregate income. 95% confidence intervals (shaded areas) are computed from re-estimating our model on 100 bootstrapped samples from our coin hoard data. The vertical lines correspond to the Arab conquests, starting in 622 under Muhammad and ending in 713 with the conquest of Iberia.

than the average region of the ancient western world. This relative economic decline of the Mediterranean starts more than a century before the birth of Islam, and therefore cannot be attributed to a newly formed religious border. It is first fueled by the relative rise of eastern regions — both before and after the birth of Islam — and then by the relative rise of northern Europe — starting in the 6th century and accelerating in the 8th century.

Our granular estimates also reveal that the relative economic decline of the Mediterranean conceals a substantial reallocation of economic activity not just away from, but also within the Mediterranean. Figure 9b shows the evolution of economic concentration from the 4th to the 9th century (Herfindahl index). Starting as early as the beginning of the 5th century, which corresponds to the split of the Roman empire between western and eastern provinces, the concentration of economic activity in the entire ancient western world falls. This fall pre-dates both the birth of Islam and the collapse of north-south Mediterranean trade, and the economic decline of the Mediterranean by more than a century. It is initially driven by a relative decline of the regions at the core of the Roman empire, northern Italy, western Europe, and Egypt, and subsequently by a relative rise of middle-eastern and eastern regions. Interestingly, while we document almost continuous shifts in relative economic activity throughout our sample period, concentration remains relatively stable from the 6th to the 9th century.

Our evidence suggests that (regional) inequality may not fall solely following catastrophic violent events (Scheidel, 2017), and that the ancient western world remained (economically) multi-polar throughout Late Antiquity (Wickham, 2006, McCormick, 2001).

5.4. Reversal of fortunes: Byzantium versus northern Europe

To showcase the promise of our structural estimates of granular real consumption time series, figure 10 displays the evolution of real consumption, for each 20-year period over AD 380-880, for Byzantium and for Francia and Germania. We highlight those two regions because they undergo some of the most striking reversal of fortune over our sample period. We briefly discuss the other regions at the end of this section.

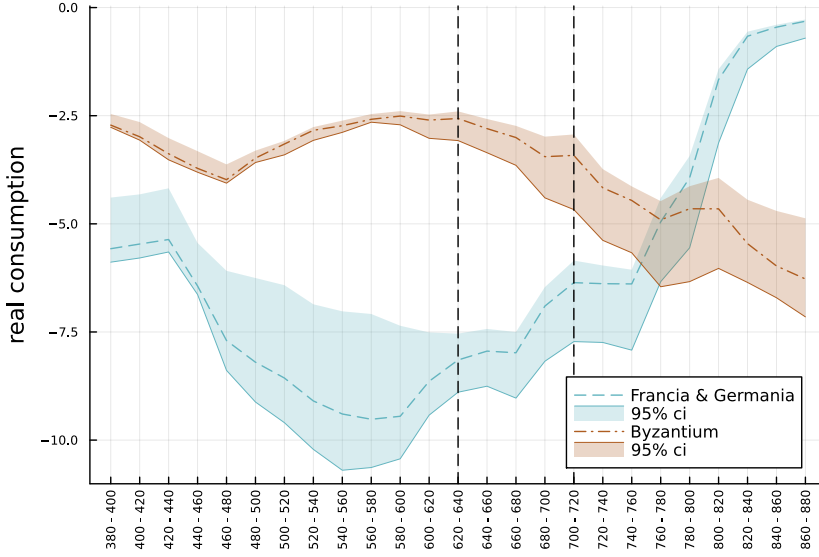


FIGURE 10.—Byzantine Heartlands versus Francia & Germania, AD 380-880. *Notes:* This figure shows time-series of (log) real consumption as a share of aggregate consumption from AD 380 to AD 880 for the Byzantine Heartlands and for Francia & Germania. See section 5.1 for details on the computation of real consumption using estimates from the specification in column 2 of table III. 95% confidence intervals (shaded areas) are computed from re-estimating our model on 100 bootstrapped samples from our coin hoard data. The vertical lines correspond to the Arab conquests, starting in 622 under Muhammad and ending in 713 with the conquest of Iberia.

For over two centuries, from the late 4th to the early 7th century, the heartlands of the Byzantine empire are stable and one of the wealthiest region of the ancient western world. Byzantium is also unique in that its real consumption is in part sustained by a very large openness to trade — the term $\pi_{nn}^{-1/\theta}$ in equation (23) — and by a large trade deficit financed by excess minting — the term $(1 - \lambda)w_n[t - 1]L_n[t - 1] + M_n[t]/w_n L_n[t]$. By contrast, the northwest European region of Francia and Germania experiences a steep decline, at a time which coincides with the fall of the western Roman empire.

Starting in the early 7th century, the fortunes of those two regions are radically altered. Byzantium undergoes a gradual economic collapse, the timing of which corresponds approximately to the Arab conquests of the lands to its east and south. This collapse is in part driven by a collapse of Byzantium’s access to international trade following the rapid decline of Mediterranean trade (see section 5.2), in part by a decline in trade deficits financed by excess minting, and in part by a fall in measured production capacity. The northwest European Frankish lands of Francia and Germania on the other hand experience a spectacular and accelerating growth in real consumption relative to other regions, becoming one of the wealthiest regions of the ancient western world over the course of the 8th century. While our estimates are solely descriptive and we do not claim any causal link, we note that the economic rise of northwest Europe happens to accelerate after Islam has reached half of the Mediterranean, in line with the thesis by Pirenne (1927) that “(...) without Mohammed Charlemagne would have been inconceivable.”

We briefly note that our estimates of economic activity for all 13 regions (see Supplemental Appendix figure B.6) have a detectable correlation with known historical events. For instance, real consumption starts falling in the 4th century in the regions at the heart of the western Roman empire, northern Italy and Aquitaine, under pressure from political instability and Germanic incursions. Egypt gradually declines as it loses its central role in the Roman economic

system. By contrast, real consumption reaches a peak in al-Sham, greater Syria, under the Umayyad caliphate of Damascus in the early 8th century; in al-Andalus under the Umayyad Caliphate of Cordoba in the late 8th century; in al-Iraq under the Abbasid caliphate of Baghdad in the late 8th century; and in the Arabian peninsula, the birthplace of Islam, in the 8th century.

We also note, starting in the 8th century, the rapid and spectacular economic rise of Atlantic regions — al-Maghrib, al-Andalus, Aquitaine and Basque country, and Francia and Germania. This evidence suggests that the rise of western Europe may have been initiated more than seven centuries before the European colonial expansion into the Americas, Africa, and Asia.

5.5. Sensitivity analysis

We conclude this section with an exploration of the sensitivity of our real consumption estimates. We consider in table IV both alternative treatments of our data, and alternative modeling assumptions. To more easily compare different series of estimates across alternatives, we solve for steady state real consumption for the AD 460-620 and the AD 700-900 periods — before and after the Arab conquests — and present regional log changes for each alternative.

Column 1 presents our baseline estimates. We use the trade costs specification from column 2 in table III; we average our estimates for bilateral and seller terms ($d_{ni}^{-\theta}[t]$ and $\tilde{\beta}_i[t]$) and for minting ($M_i[t]$) separately for AD 460-620 and for AD 700-900, and we follow the steps in section 5.1 to compute steady state real consumption relative to aggregate consumption for each region and period. For each region, we compute the log change in relative real consumption from AD 460-620 to AD 700-900. A positive number means that a region is getting larger as a share of the ancient western world.⁴³ Column 1 confirms the stylized facts discussed in section 5.4, for instance the reversal of fortune between Byzantium versus Francia and Germania, the rise of Islamic al-Andalus and Carolingian Aquitaine, or the decline of Italy and Egypt.

Columns 2, 3, and 4 use different treatments of our coin data. Column 2 uses solely gold and silver coins — about 25% of our dataset — and computes coin values in equivalent grams of gold instead of counts, using known exchange rates and denominations. Columns 3 and 4 aggregate our coin dataset into shorter 10-year intervals (column 3) or longer 30-year intervals (column 4) than our 20-year intervals baseline. After re-estimating all parameters from the specification of column 2 in table III for each dataset, we recover real consumption series using the same model as in column 1. Columns 5 and 6 instead use the same parameters as column 1, but a different model. In column 5, we set the trade elasticity θ to 12 instead of 4 as in our baseline. In column 6, we assume a simple model with endogenous monetization.⁴⁴

Across all specifications, while the levels of consumption changes vary, they are highly correlated with our baseline, and the qualitative patterns of relative changes are similar. The correlations with our column 1 baseline ranges from 65% to 99.95% for log changes, and from 45% to 99% for the ranks of log changes. The correlation is lowest for the small sample of gold and silver coins in column 2. We confirm across all specifications the reversal of fortunes between the heartlands of the Byzantine empire and the Frankish northern European region of Francia and Germania, the rise of al-Andalus and Aquitaine, and the decline of Italy.

⁴³If the world is shrinking, a positive number means this region is either growing or shrinking at a slower pace than the rest. The values reported in table IV are equivalent to projecting log consumption series on region and period fixed effects: expressing regional consumption relative to global consumption removes period fixed effects (in logs), and computing (log) differences between periods removes time-invariant region fixed effects.

⁴⁴We present in Supplemental Appendix C a simple extension of our model with barter and monetized transactions inspired by Kiyotaki and Wright (1989). We explain how to estimate this extended model from coin data. In practice we simply raise the term $T_n^{1/\theta} L_n$ in equation (23) to the power $1/1 + \theta$ to account for endogenous monetization.

TABLE IV
REAL CONSUMPTION FROM AD 460-620 TO AD 700-900: SENSITIVITY ANALYSIS

	(1)	(2)	(3)	(4)	(5)	(6)
Real consumption change: $\Delta \log (X_n/p_n)$						
al-Andalus (Iberia)	3.20	0.73	3.35	3.44	2.91	0.55
Aquitaine, Basque Country	6.89	4.99	7.17	4.92	6.45	1.36
Francia, Germania	10.16	9.48	8.48	8.70	9.35	2.05
Northern Italy, Balkans	-1.23	-2.84	-1.05	-1.88	-1.04	-0.31
Southern Italy	-2.62	-1.81	-5.24	-1.88	-2.33	-0.45
Byzantine Heartlands	-3.82	-2.02	-4.05	-4.45	-3.50	-1.47
al-Sham (Greater Syria)	-1.20	-0.25	-1.93	-5.04	-1.26	-0.42
Northern Syria, Caucasus	0.30	2.49	1.60	0.33	0.18	0.12
al-Iraq, al-Jibal, ...	0.15	-1.09	-0.60	0.72	-0.05	-0.05
Eastern Caliphate	1.20	-3.48	-2.01	0.10	1.14	0.19
Jazirat al-arab, al-Yaman	3.07	-6.68	2.15	2.23	2.82	0.80
Misr (Egypt)	-4.33	-1.72	-1.69	-3.68	-3.96	0.18
al-Maghrib	-0.52	1.72	1.49	1.35	-0.31	-0.09
Correlation with baseline (column 1)						
Pearson corr.		0.65	0.92	0.94	1.00	0.91
[95% CI]		[0.39 , 0.88]	[0.78 , 0.93]	[0.80 , 0.96]	[0.84 , 1.00]	[0.74 , 0.92]
Spearman rank corr.		0.45	0.82	0.88	0.99	0.84
[95% CI]		[0.26 , 0.75]	[0.63 , 0.90]	[0.64 , 0.93]	[0.75 , 1.00]	[0.53 , 0.87]
Period length	20 years	20 years	10 years	30 years	20 years	20 years
Coin Accounting	Number	Value	Number	Number	Number	Number
Sample	All	Gold/Silver	All	All	All	All
Model	Standard	Standard	Standard	Standard	$\theta = 12$	w. Barter

Note: This table shows alternative estimates of relative changes in real consumption between AD 460-620 and 700-900. Column 1 uses our baseline specification corresponding to column 2 of table III; we compute regional real consumption in two steady state equilibria from parameters averaged over AD 460-620 and 700-900, and report log changes. Column 2 uses gold and silver coin values instead of counts (in equivalent grams of gold, assuming a constant exchange rate of 12g of silver for 1g of gold). Columns 3 and 4 use our baseline specification but aggregates our data into 10-year or 30-year intervals, adjusting the coin loss rate to the period length (always 1.7% p.a.). Column 5 uses our baseline estimates but sets the trade elasticity to $\theta = 12$. Column 6 uses an extension of our baseline model with both monetized and barter transactions inspired by Kiyotaki and Wright (1989) — see Supplemental Appendix C for details. The bottom panel shows Pearson and Spearman correlations with our baseline in column 1. 95% confidence intervals, in square brackets, are computed from re-estimating our model on 100 bootstrapped samples from our coin hoard data.

Interestingly, our estimates for real consumption are stable despite the fact that the estimated trade cost parameters vary as we modify our dataset across columns 1-4 (see Supplemental Appendix table B.VIII): the estimated travel time elasticity of trade is lower when using value-weighted gold and silver coins, and when aggregating our data into longer time intervals, as expected from the discussion in section 3.2. We also note that even tripling the trade elasticity θ from 4 to 12 in column 5 barely affects our estimates for real consumption changes.

Finally, we observe that allowing for an endogenous level of monetization in column 6, where wealthier regions rely proportionately more on monetized exchanges, does affect the *absolute* levels of real consumption changes, but it mostly preserves their *relative* levels. This specification relies on the plausible assumption that as regions develop, more transactions become monetized. While we are not aware of any systematic information on the degree of monetization which would allow us to validate our simple model of endogenous monetization, we believe the estimates reported in column 6 to be our most credible.

TABLE V
ACTIVE MINES AND MINT OUTPUT

	Dep. var.: Log Mint Output M_{it}		
	(1)	(2)	(3)
$(\text{Active Mine})_{it}$	1.475 (0.35)	1.724 (0.34)	1.332 (0.46)
20-year time period FE	No	Yes	Yes
Region FE	No	No	Yes
R^2	0.0577	0.306	0.433
Observations	343	343	343

Note: The dependent variable is logged minting, $\log(M_i[t])$, estimated in section 4.3. The independent variable is a dummy equal 1 iff region i has a documented active mine at the start of a 20-year period t . Standard errors in parentheses, clustered at the 20-year period level.

6. ANCIENT MINES, CLIMATE, CITIES, BATTLES, AND PLAGUES

In this section, we confront our estimates with several external data sources. Our minting estimates are strongly correlated with the known exploitation of ancient mineral deposits (section 6.1). Our real consumption estimates are strongly correlated with paleo-climatic shocks (section 6.2) and with urbanization (section 6.3), but not with military conflicts or the Justinianic plague of the 6th century (section 6.4). The evidence in this section is not meant to capture causal effects — except for paleo-climate. Instead, it serves both as a validation of our estimates and as an informal test of the determinants of ancient economic activity.

6.1. Ancient mines and minting output

We collect information on 37 known ancient mines, their geographic location, approximate dates of operation, and the metal being mined — gold, silver, and copper — from the scholarly literature in history and archaeology (see Supplementary Appendix A.5). We regress (logged) minting output on a dummy variable equal to one if region i has a documented active mine at the start of a 20-year period t .

The results are presented in table V. It shows a large and statistically significant association between ancient mining activity and our minting estimates, robust to controlling or not for region and period fixed effects. The presence of an active mine is associated with a four-fold increase in minting output ($e^{1.3} \approx 3.67$ in column 3).

This association is not causal, as the decision to exploit a mine is endogenous and may be driven by the desire of local rulers to increase minting output. However, to the extent that the presence of accessible mineral deposits is exogenous to local monetary policy, it suggests that a lower cost of minting leads to a higher minting output. This is consistent with our model, where an increase in minting in one region, everything else equal, increases real consumption. Minting in one region generates inflation, but since inflation is partly exported, the private (regional) gain from minting is always strictly positive. The equilibrium of a minting game — which we do not explicitly model — would be such that the gain from minting equals its cost; with an increasing and convex cost, an easier access to mineral deposits induces more minting.

6.2. Paleo-climate and real consumption

We show evidence that real consumption is strongly affected by adverse climate shocks. We collect granular data on summer — June to August — drought severity indices reconstructed

TABLE VI
CLIMATE SHOCKS AND REAL CONSUMPTION

	Dep. var.: Log Real Consumption X_n/p_n			
	(1)	(2)	(3)	(4)
# Years with medium drought	-0.0578 (0.070)			
# Pairs of consecutive years with medium drought		-0.260 (0.13)		
# Years with severe drought			-0.380 (0.21)	
# Pairs of consecutive years with severe drought				-1.234 (0.50)
20-year time period FE	Yes	Yes	Yes	Yes
Region FE	Yes	Yes	Yes	Yes
R^2	0.322	0.327	0.328	0.333
Observations	399	399	399	399

Note: The dependent variable is logged real consumption, $\log(X_n[t]/p_n[t])$, estimated in section 5.1. The independent variables are measures of drought. For each region and each year, we average the values of the Palmer Drought Severity Index (PDSI) over a four-by-four degree rectangle centered on the node in our transport network nearest to the count-weighted mean of mint locations in the region. The PDSI is a normalized index of wet/dry conditions, taking values between -10 and +10, with negative values corresponding to dry conditions. Our drought measures are: in column 1, the number of years of moderate drought (index below -2) for region n and 20-year period t ; in column 2 the number of pairs of consecutive years of moderate drought; in column 3 the number of years of severe drought (index below -3); in column 4 the number of pairs of consecutive years of severe drought. Standard errors in parentheses, clustered at the 20-year time interval level.

from tree-rings chronologies from the Great Eurasian Drought Atlas (Burnette, 2021).⁴⁵ We define four variables: the number of years of moderate or severe drought for region n and 20-year period t , and the number of pairs of consecutive years with moderate or severe droughts. Those climate shock measures vary both over time and space, allowing us to isolate their impact on our real consumption series controlling for region and period fixed effects. This is crucial because our estimation strategy cannot identify aggregate variation over time, and because the definition of regions is to some degree arbitrary.

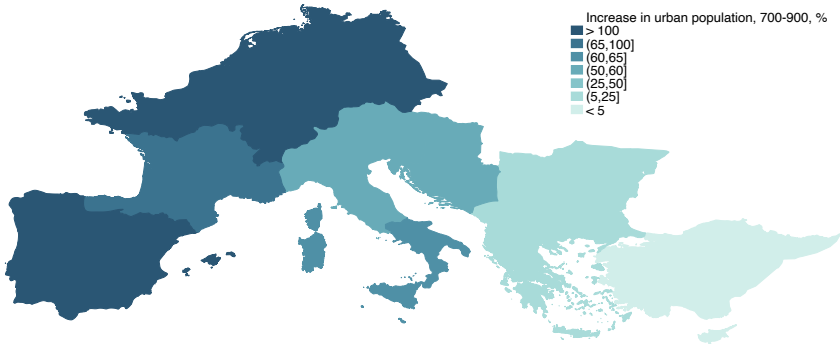
Table VI shows the results of regressions of (logged) real consumption on droughts. We find a large and significant association between droughts and real consumption. To the extent that severe drought events are plausibly exogenous, we can interpret those coefficients as causal. Column 3 suggests that one year of severe drought reduces real consumption by 32% ($1 - e^{-0.38} \approx 0.32$). Using the estimates from a model with endogenous monetization, one year of severe drought causes a 7% reduction in real consumption (treatment $\approx 1 - e^{-0.38/5} \approx 0.07$).

Those results show that weather is an important determinants of aggregate productivity, production, and ultimately real consumption. Since our estimates for real consumption are derived from coin and trade flow data, those results also suggest either that agricultural products—plausibly affected by droughts—were traded, and/or that non-farm industrial productivity was also negatively affected by droughts. Our results extend by more than one millennium the finding from Dell, Jones, and Olken (2012) that adverse climate shocks negatively affect both agricultural and industrial productivity among poor countries in the late 20th century.

⁴⁵See <http://drought.memphis.edu/GEDA>.



(a) Relative real consumption changes: pre- to post-Arab conquests



(b) Urban Population Growth in European Regions, AD 700-900

FIGURE 11.—Real consumption and urbanization. *Notes:* Panel (a) shows the logged relative real consumption change from the pre- to the post-Arab conquests period, column 1 of Table IV. Panel (b) shows the percentage growth of the urban population post-Arab conquests, between AD 700 and AD 900. City size data from Buringh (2021), Bosker, Buringh, and Van Zanden (2014), Buringh (2020), except for Byzantine Anatolia, which is not covered. We construct measures of urban decline in Anatolia based on the shrinking surface area of cities described in Brandes (1989). The resulting figure of a 10% decline in urban population over this period seems to be a conservative estimate given that many coastal cities saw large amounts of destruction and depopulation as a consequence of Arab raids.

6.3. Urbanization in ancient Europe

We confront our estimates for changes in real consumption to realized changes in urbanization in Europe. While our model does not feature any explicit notion of urbanization, we conjecture that a higher real consumption allows to sustain a larger urban population. This exercise is illustrative, meant to verify that our estimates for real consumption derived solely from information on coin flows are consistent with independent evidence on economic growth.

Figure 11 shows our estimates for changes in real consumption (top panel, 11a) together with urban population growth north of the Mediterranean over AD 700-900, aggregated from city size data (bottom panel, 11b).⁴⁶ Comparing both maps suggests that our estimates for real consumption are qualitatively in line with independent evidence on urbanization. Our estimated drop in real consumption in the heartlands of the Byzantine empire, and substantial increase in western and northern Europe are consistent with urban population declines in Asia Minor and

⁴⁶While ancient city size estimates are naturally imprecise, panel 11b is in line with the consensus view of increased urbanization in northwestern Europe, and stagnation or decline in the eastern Mediterranean.

Cyprus, low urban growth in Greece, Thracia, and Dacia, medium urban growth in the Balkans, and Italy, and strong urban growth in Iberia, Aquitaine, and Francia/Germania.

6.4. Additional results

Real consumption is not correlated with military conflicts. We find no systematic association between instances of military conflict and our estimates for real consumption (see Supplemental Appendix B.10 and table B.IX). We collect data on 297 major battles and sieges for AD 300-950 as covered by [Jaques \(2006\)](#). Real consumption in region n and period t is correlated neither with the number of battles in the region-period, nor with having at least one recorded battle, even controlling for region and period fixed effects. This non-result may reflect countervailing forces, where on the one hand wealthier regions are better able to mount military campaigns, while on the other violence disrupts production and trade. It may also be due to the selective and biased recording of violent confrontations in the literature.

Real consumption is not correlated with the Justinianic plague. We do not find any systematic association between the local incidence of the Justinianic plague ([Stathakopoulos, 2004](#)) and real consumption (see Supplemental Appendix B.10 and table B.X). This non-result may be due to the fact that the plague was truly a global event and had a uniform impact across regions, which our estimates are unable to detect since we can only recover relative consumption each period; or the documented differences in lethality across regions may only reflect differences in the availability or quality of literary sources; or it may be consistent with the hypothesis that the Justinianic plague was after all an ‘inconsequential pandemic’ ([MENI+, 2019](#)).

7. CONCLUSION

We study the patterns of change in economic geography around the Mediterranean during Late Antiquity through the lens of coin flows. We propose a dynamic model of trade where coins are used for transactions, so that they gradually diffuse over space and time in proportion to trade flows. We estimate this model using data on the composition of ancient coin finds. We then use these parameters to recover granular time series for trade and relative real consumption from the fourth to the tenth century. Our estimates for changes in real consumption are consistent with measures of relative urbanization across European regions. They also suggest that climate shocks were key drivers of real consumption, while military conflicts were not.

Our evidence from coin finds indicate that north-south Mediterranean trade flows declined following the Arab conquests. This can be explained by a newly formed trade barrier between the emerging Islamic Caliphate and the Christian West. These changes in trade patterns are in line with the claims of a trade disruption in the Mediterranean by [Pirenne \(1939\)](#). Our results show, however, that intra-caliphal trade in the Mediterranean remained vibrant. Moreover, Pirenne believed that trade disruptions also led to a vast reduction in economic activity and exchange within the Frankish lands. We find instead that Frankish regions experienced a spectacular economic rise over that period, contrasting the relative economic decline of the Byzantine empire. Our estimates also reveal that the start of the relative decline of the Mediterranean predates the birth of Islam, contradicting mono-causal theories of change during Late Antiquity that attribute the entirety of the shift in economic activity to the Arab conquests.

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